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Nuclear export governs TDP-43 phase transitions and cytoplasmic aggregation

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eLife Assessment

This study presents **valuable** findings implicating nuclear export in the regulation of protein condensate behaviour and TDP-43 phase behaviour, suggesting a link to pathogenic aggregation in ALS/FTD. However, the research relies extensively on synthetic, non-physiological protein variants and a homozygous disease model, with limited mechanistic validation, leading to conclusions that remain largely correlative. Furthermore, limitations in the reporting of experimental replication and controls, as well as inconsistencies between cancer cell and neuronal models, diminish confidence in the robustness of the findings. Despite its technical strengths, the findings presented are currently **incomplete** and do not provide sufficient evidence to substantiate claims about the direct role of nuclear export in pathological protein aggregation and disease.

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Abstract

RNA-binding protein TAR DNA-binding protein 43 (TDP-43) can form liquid-like, nuclear assemblies whose phase transitions are linked to protein aggregation and neurodegeneration. The mechanisms that determine whether these assemblies remain dynamic or convert into solid states are poorly defined. Here we combined chemical and genome-wide genetic screenings to identify cellular processes influencing the phase behavior of an RNA-binding defective TDP-43 mutant associated with amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD). While the screens identified multiple cellular processes—including RNA splicing, translation, and proteostasis—as modulators of TDP-43 phase states, our analyses underscore nuclear transport as a critical determinant of liquid-to-solid transitions. Inhibition of nuclear export prevents cytoplasmic accumulation of immobile TDP-43 assemblies, whereas enhanced export promotes irreversible, solid TDP-43 structures. We validated this mechanism in a brain organoid model of ALS, showing nuclear retention limits pathogenic aggregation and its associated defects. These findings establish nuclear export as a key regulator of TDP-43 phase transitions and define a mechanistic link between altered phase dynamics and neurodegeneration.

Introduction

The RNA-binding protein TAR DNA-binding protein 43 (TDP-43), encoded by the *TARDBP* gene, plays a central role in RNA transport, splicing and metabolism (Guo and Shorter, 2017 [↗](#); Suk and Rousseaux, 2020 [↗](#)), and is the major pathological component of cytoplasmic inclusions in amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD) patients (Kellett et al., 2025 [↗](#); Neumann et al., 2006 [↗](#); Rummens and Da Cruz, 2025 [↗](#)). TDP-43 contains two RNA-

recognition motifs (RRMs), a bipartite nuclear-localization sequence, and a glycine-rich C-terminal low-complexity domain (LCD) that is intrinsically prone to aggregation (Tziortzouda et al., 2021). Under physiological conditions, TDP-43 binds RNAs in the nucleus, participating in multiple RNA-metabolic processes. In contrast, disease-causing mutations trigger its mis-localization to the cytoplasm, forming detergent-resistant inclusions. TDP-43-positive protein aggregates were present in ~97% of ALS and ~45% of FTD cases, as well as in a subset of Alzheimer's disease patients. These diseases are now collectively termed TDP-43 proteinopathies (Brettschneider et al., 2014; Neumann et al., 2006). Both loss of nuclear function and gain of cytoplasmic toxicity have been implicated in neurodegeneration (Suk and Rousseaux, 2020; Tziortzouda et al., 2021), underscoring the importance of nucleocytoplasmic transport in regulating TDP-43 pathophysiological functions.

Like many RNA-binding proteins (RBPs), TDP-43 undergoes liquid-liquid phase separation to form membraneless biomolecular condensates (Conicella et al., 2016; Shorter, 2019; Tziortzouda et al., 2021). Under conditions of cellular stress or in the presence of RNA-binding defective mutations, these condensates can transition from dynamic liquid droplets to more rigid gel-like or solid assemblies (Conicella et al., 2016; Gasset-Rosa et al., 2019; Lang et al., 2024). To date, more than 70 disease-linked mutations in *TARDBP* have been identified in ALS or FTD, many within the LCD, while others clustering near or in the RRM domains disrupt RNA binding (Tziortzouda et al., 2021). Defects in RNA binding appears to be a major governor of TDP-43 phase behavior (Babinchak et al., 2019; Mann et al., 2019). Additionally, disease-associated mutations alter TDP-43's regulatory functions in RNA metabolism, causing widespread cryptic exon inclusion (Ling et al., 2015; Ma et al., 2022; Mehta et al., 2023; Seddighi et al., 2024; Zhang et al., 2025b), alternative polyadenylation and other RNA maturation defects that collectively contribute to neurotoxicity (Zeng et al., 2025).

A major advance in the field of TDP-43 phase regulation is the recent report of a unique form of TDP-43 demixing, forming "anisosomes" that contain an anisotropic spherical shell of an RNA-binding-defective TDP-43 mutant and a central liquid core formed by the HSP70 chaperone (Yu et al., 2021). These structures, observed mostly in the nucleus, were postulated to be an intermediate en route to pathological aggregation when cellular proteostasis or nucleocytoplasmic transport is compromised. Additionally, *in vitro* studies with purified TDP-43 and selected small molecules have begun to elucidate the molecular basis of TDP-43 self-association (Maurel et al., 2025; Rubien et al., 2022), offering insights into its phase behavior and aggregation mechanisms.

Despite these advances, the cellular pathways modulating TDP-43 phase behavior (droplet formation, anisosomal shell-core architecture, conversion to gel/solid structures) and its link to disease-associated aggregation remain poorly defined. Here, we address this knowledge gap by combining a chemical-genetic screen approach with a genome-wide siRNA screen to identify molecular determinants that modulate the phase behavior of an RNA-binding defective TDP-43 mutant. Our study reveals critical contributions of RNA splicing, protein translation, and the HSP90/ubiquitin-proteasome proteostasis network to TDP-43 phase transitions. Importantly, our work demonstrates how nuclear export can influence the transition of TDP-43 from a nuclear liquid-demixed form to a cytoplasmic immobile structure. We validated the impact of nuclear export inhibition on phosphorylated-TDP-43 cytoplasmic accumulation using an iPSC-derived 3-D organoid model bearing an ALS-associated mutation (Zhang et al., 2025b). Collectively, these findings establish a mechanistic framework linking altered phase dynamics and TDP-43 aggregation to nuclear transport defects, a process known to modulate neurodegeneration in ALS and FTD.

Results

A chemical genetic screen identified modulators of TDP-43 phase behavior

To identify cellular pathways modulating TDP-43 phase behavior, we conducted a chemical genetic screen using a small molecule library targeting diverse known cellular pathways and processes. We employed a previously established DLD1 cell model stably expressing an RNA-binding defective TDP-43 mutant (2KQ) tagged with the green fluorescence protein Clover because TDP-43 2KQ undergoes rapid demixing in the nucleus upon expression induction (Yu et al., 2021 [4](#)). We seeded cells in 384-well plates and induced the expression of TDP-43 2KQ with doxycycline for 24 hours. We then treated these cells with a LOPAC library containing 1280 drugs for 24 hours (Figure 1A [4](#)). This approach could avoid false positive drugs that affect the induction of TDP-43 2KQ.

High content imaging combined with algorithm-based image analyses (see methods) detected ~20 spherical anisosomes per cell in controls and most treated cells. However, in cells treated with a subset of the drugs, the number of anisosomes was noticeably reduced (Supplemental Table 1 [4](#)). Importantly, cells treated with Bortezomib, a potent proteasome inhibitor, had fewer but enlarged TDP-43 positive punctae with irregular shapes (Figure 1B [4](#)), similar to cells treated with another proteasome inhibitor reported previously (van Eersel et al., 2011 [4](#); Yu et al., 2021 [4](#)). This result validated our screening design.

To further confirm the identified hits, we performed a titration experiment, which showed that these inhibitors dose-dependently reduced the number of anisosomes with IC_{50} ranging from sub-micromolar to micromolar (Figure 1B [4](#), Supplemental Table 1 [4](#)). Interestingly, the screen identified cytoplasmic signaling kinases including PAK4, BTK, and GSK-3 β as well as the receptor tyrosine kinase PDGFR β as modulators of TDP-43 phase behavior (Figure 1B [4](#), Supplemental Table 1 [4](#)). PAK4 and GSK-3 β are known regulators of the YAP signaling pathway (Jin et al., 2025 [4](#); Mo et al., 2024 [4](#)), and YAP was recently identified as a TDP-43-interacting protein that influences its phase dynamics (Zhang et al., 2025a [4](#)). These connections provide additional physiological support for the relevance of the modulators uncovered in our screen. Our screen also linked TDP-43 2KQ demixing to protein translation because cycloheximide, a ribosome elongation inhibitor could modulate TDP-43 phase dynamics (see below).

Live cell imaging revealed two distinct types of TDP-43 phase modulators

We next used confocal microscopy to further characterize the impact of a subset of the identified inhibitors on anisosome dynamics. We chose Tripterin (a drug targeting the heat shock protein HSP90), CP-673451 (a PDGFR β inhibitor), two deubiquitinase inhibitors VLX-1570 and Spautin-1, and the Exportin-1 inhibitor KPT-276 because previous reports have linked the targets of these compounds to TDP-43 proteinopathies (Archbold et al., 2018 [4](#); Farrawell et al., 2020 [4](#); Furukawa et al., 2015 [4](#); Garcia-Toscano et al., 2024 [4](#)). We first induced anisosome formation for 24 hours and then treated cells with the inhibitors. Compared to DMSO-treated controls, cells treated with Spautin-1 had fewer and smaller anisosomes. KPT-276 treatment also reduced anisosome number, albeit to a lesser extent (Figure 2A [4](#), top panels). In contrast, cells treated with CP-673451, Tripterin, or Bortezomib had fewer but larger TDP-43 punctae. Since Tripterin was also reported as a 20S proteasome inhibitor, we confirmed the role of HSP90 in this process by treating cells with the well-established HSP90 inhibitor Geldanamycin, which caused similar phenotypes as Tripterin (Supplemental Figure 1A [4](#)). These results suggest that TDP-43 phase dynamics are modulated by the cellular folding capacity, the ubiquitin proteasome system, and nucleocytoplasmic transport.

To further dissect the effects of these inhibitors on TDP-43 phase separation, we combined drug treatment with live cell fluorescence imaging, using Fluorescence Recovery After Photobleaching (FRAP) or reversed FRAP to define the dynamics of TDP-43 2KQ. In FRAP, when the green

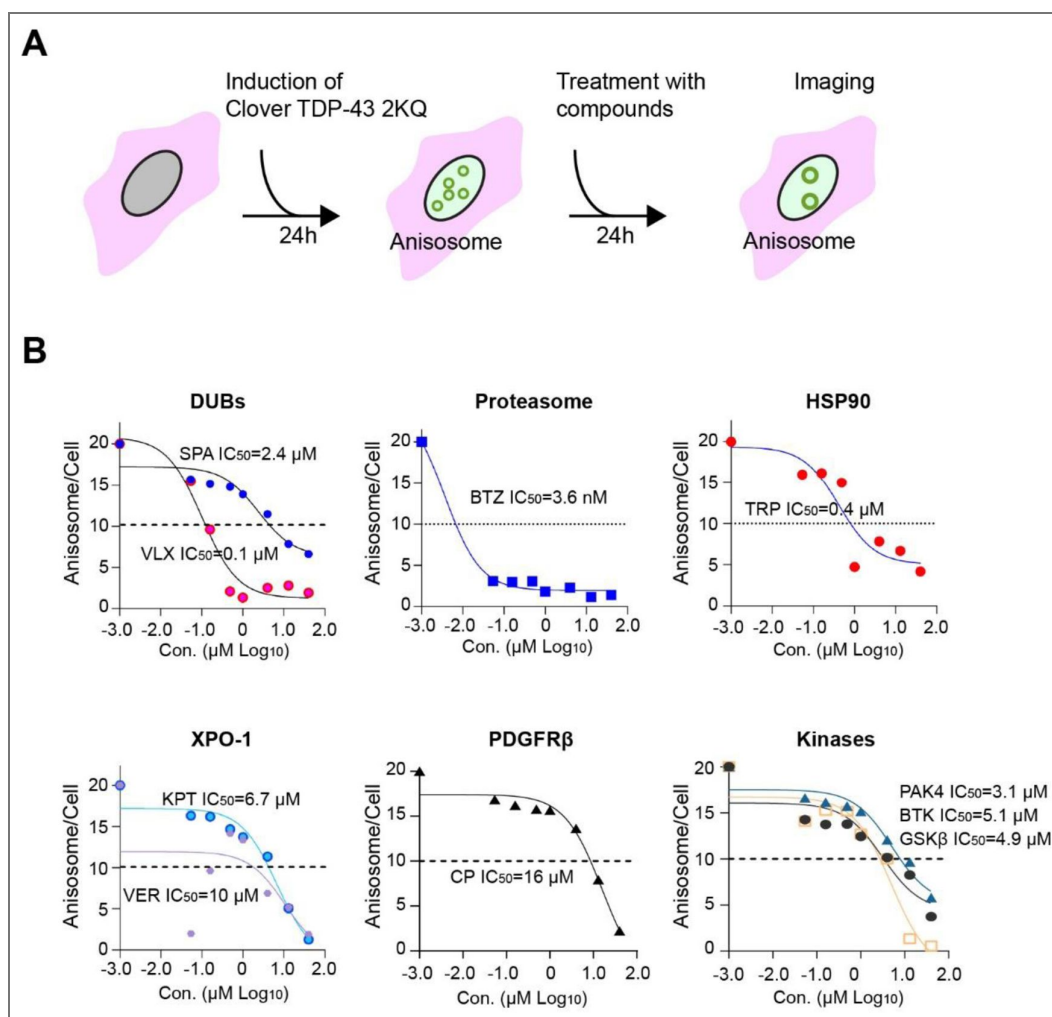


Figure 1. A chemical genetic screen identifies TDP-43 phase modulators

(A) The workflow of the chemical genetic screen. **(B)** Dose dependent reduction of anisosome number by identified chemicals. DLD1 TDP-43 2KQ-Clover cells were treated with doxycycline to induce anisosome for 24 h and then treated with the drugs as indicated for 15 h. Cells were imaged for anisosome count.

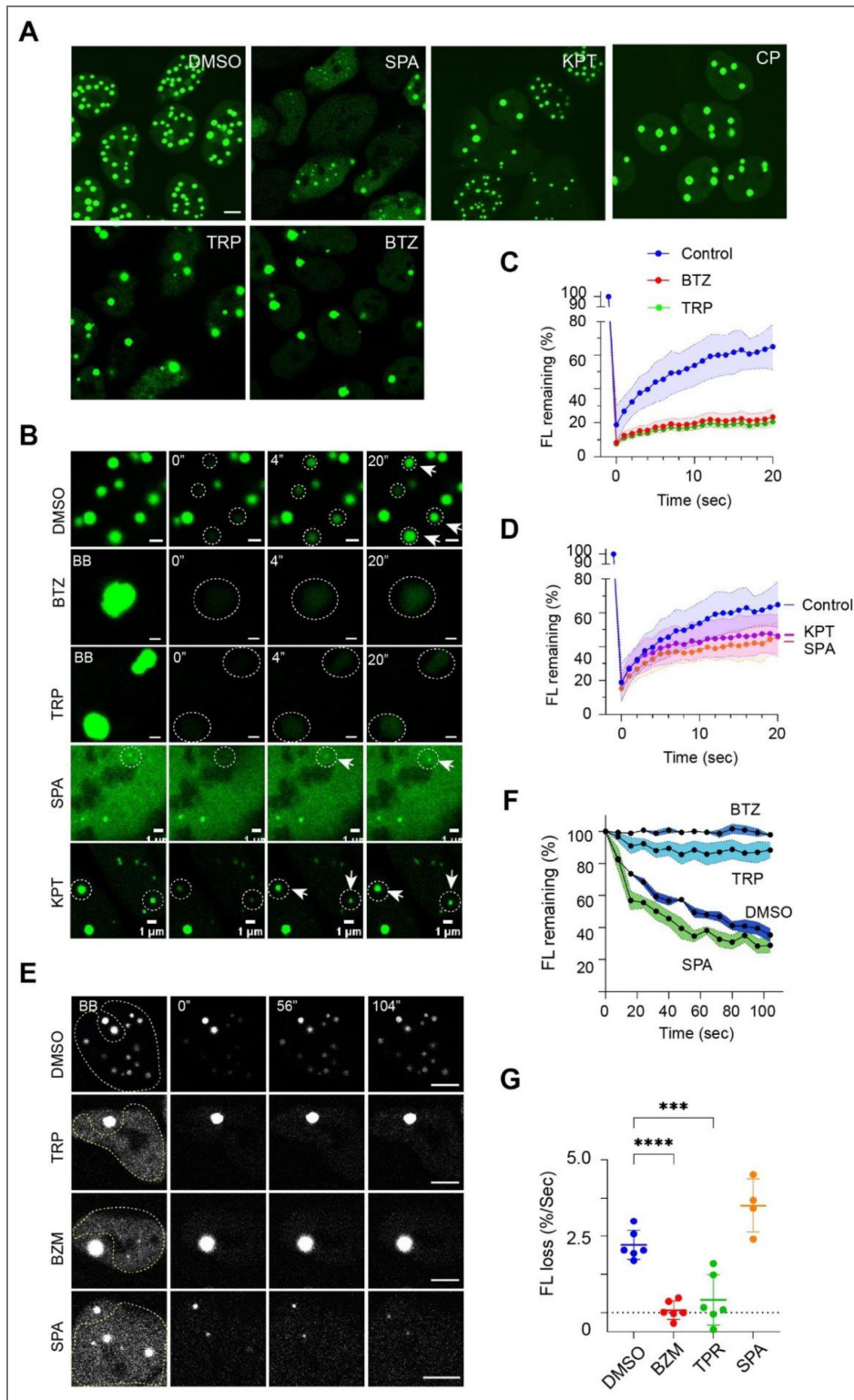


Figure 2. Two distinct types of TDP-43 phase modulators

(A) DLD1 TDP-43 2KQ-Clover cells treated with doxycycline for 24 h were further treated with the indicated compounds for 6 h and imaged (SPA, Spautin-1, 5 μ M; KPT, KPT-276, 15 μ M; CP, CP-673451, 30 μ M; TRP, Tripterin, 1 μ M; BTZ, Bortezomib, 10 nM). Scale bar, 5 μ m. (B) FRAP analyses of anisomes after treatment with the indicated drugs for 3–5 h. Circles indicate bleached areas. Scale bars, 1 μ m. (C, D) Quantification of the experiments represented in B. FL, Fluorescence. (E) Reverse FRAP analyses of anisome dynamics. The areas indicated by the dashed lines were bleached. Scale bar, 5 μ m. (F) Quantification of the fluorescence loss over time in unbleached anisomes as shown in E. N=4–6. (G) Quantification of the initial rate of fluorescence loss in unbleached anisomes. ****, $p < 0.0001$; ***, $p < 0.001$ by unpaired Student's t-test.

fluorescence of anisosomes was bleached by a laser in untreated cells, it rapidly recovered due to the recruitment of TDP-43 2KQ from the nucleoplasm (Figure 2B, top panels, Figure 2C, D). Conversely, in reversed FRAP, bleaching TDP-43 2KQ surrounding an anisosome and then imaging the anisosome over time revealed a gradual decline in fluorescence intensity within the unbleached anisosome, while the surrounding bleached anisosomes regained some fluorescence (Figure 2E, top panel). This result suggests that TDP-43 2KQ in anisosomes could also be released into the nucleoplasm. Interestingly, in cells treated with Spautin-1 or KPT-276, anisosome-associated TDP-43 2KQ could freely exchange with the nucleoplasmic pool (Figure 2B, D). However, in cells treated with Bortezomib, Tripterin or Geldanamycin, the enlarged TDP-43 puncta appeared static; TDP-43 2KQ fluorescence within anisosomes failed to recover after photobleaching (Figure 2B, C, Supplemental Figure 1B), nor did it decrease over time when surrounding TDP-43 2KQ was bleached (Figure 2E-G). Collectively, these results suggest that these inhibitors affect anisosomes via two distinct mechanisms: one maintains TDP-43 in a demixed liquid state, while the other converts it to a gel-like, solid state.

A genome-wide siRNA screen identified genetic modulators of TDP-43 phase separation

To further decipher the molecular determinants of TDP-43 phase separation, we conducted an unbiased genome-wide siRNA knockdown (KD) screen (Figure 3A). To this end, we transfected a siRNA library targeting 21,404 human genes each with three siRNAs into DLD1 cells stably expressing Clover-tagged TDP-43 2KQ. After gene KD, anisosomes were induced for 24 hours. High content imaging and automated analysis identified 1,533 candidates whose KD reduced the anisosome number per cell (Z-score > 2) (Supplementary Table S2). Based on a STRING gene network analysis, we selected 211 networked genes with top Z-scores and re-screened each of them with three additional siRNAs. This confirmed the involvement of 110 genes in TDP-43 anisosome regulation (Figure 3A, Supplemental Table S3). GO pathway analyses categorized identified genes into several major pathways including RNA splicing, protein translation, proteasomal degradation, and nuclear transport (Figure 3B). The identification of large number of ribosomal proteins and proteasome subunits is consistent with our chemical genetic screen, which showed that translation and proteasome inhibitors modulate anisosomes (Supplemental Table 1). Interestingly, GO analyses using ‘molecular function’ further linked many identified genes to neurodegenerative diseases, particularly ALS (Figure 3C, D).

Anisosome dynamics is modulated by RNA splicing and protein translation

Given the well-established function of TDP-43 in RNA binding and processing, we investigated the role of RNA splicing in anisosome regulation. To this end, we first incubated Clover-tagged TDP-43 2KQ cells with doxycycline to induce anisosomes and then treated cells with the potent RNA splicing inhibitor Pladienolide-B (PlaB). Confocal microscopy confirmed that RNA splicing inhibition resulted in fewer but larger TDP-43-positive puncta in a dose dependent manner (Figure 4A, B, Supplemental Figure 2A). Despite the size difference, anisosomes in drug-treated cells were morphologically indistinguishable from controls. FRAP experiments further showed that TDP-43 within PlaB-treated condensates remained highly mobile, although the fluorescence recovery rate was slightly reduced compared to controls (Figure 4C, D). Time-lapse live cell imaging frequently detected fusion of TDP-43 puncta after PlaB treatment (Figure 4E, Movie S1). Together, these results suggest that disrupting RNA splicing stabilizes TDP-43 in demixed liquid state, resulting in larger condensates.

Next, we explored the effect of translation inhibition on anisosome dynamics using Cycloheximide (CHX) and Anisomycin (ANS). Confocal imaging showed that after anisosome induction, translational shutdown by CHX or ANS did not significantly reduce TDP-43 expression, suggesting a long half-life. Similar to PlaB-treated cells, CHX- and ANS-treated cells contained fewer, enlarged

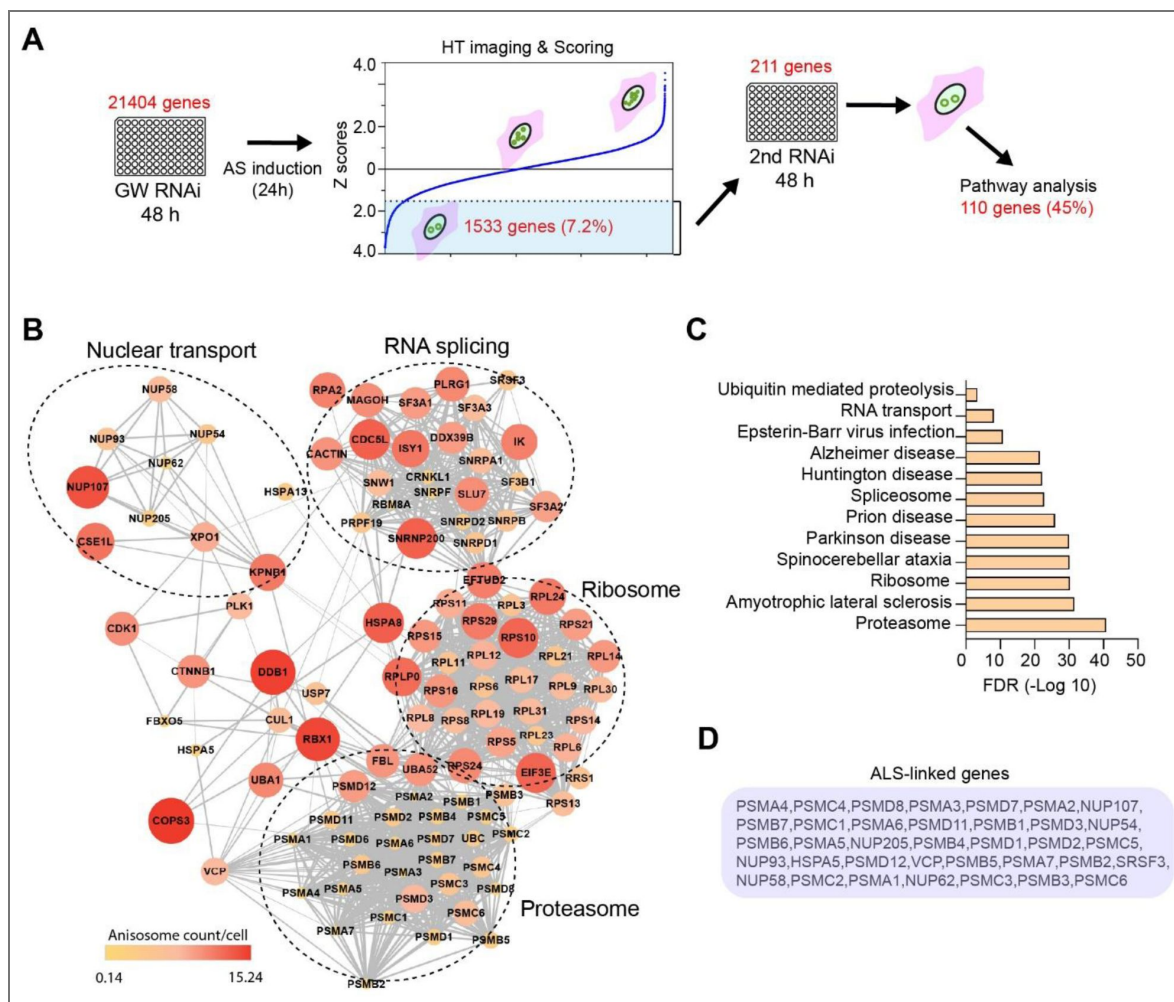


Figure 3. A genome-wide siRNA screen identifies modifiers of TDP43 phase behavior

(A) The workflow of the siRNA genetic screen. GW, genome-wide; AS, anisosome; HT, high-throughput. **(B)** Pathway analysis of genes whose knockdown reduces anisosome number in cells. The relative anisosome count are indicated by both color and size of the nodes. **(C)** GO molecular function analysis of TDP-43 phase modifiers. **(D)** A list of identified genes linked to ALS in C.

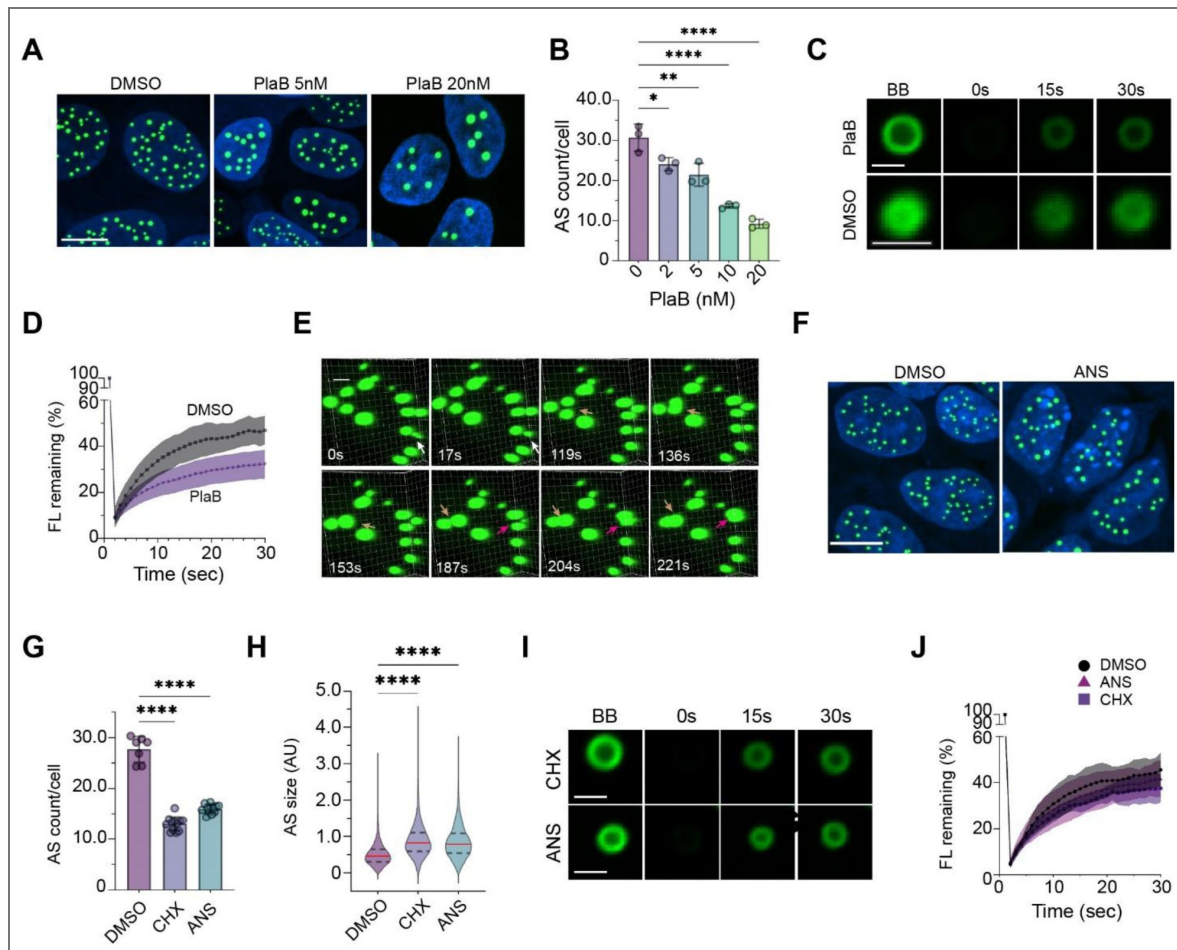


Figure 4. Anisosome phase behavior is modulated by RNA splicing and protein translation

(A) The splicing inhibitor Pladienolide-B (PlaB) reduces anisosome number in a dose dependent manner. Representative images of anisosome-induced cells treated with DMSO (control), 5 nM, or 20 nM PlaB for 16 h. Scale bar: 10 μ m. (B) Quantification showing the number of anisosome (AS) per cell in TDP-43 2KQ-Clover cells treated with PlaB as indicated. * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$ by One-way ANOVA. $n=3$ biological repeats. (C) Representative FRAP images of anisosomes in cells treated for 16 h with DMSO or PlaB (20 nM). BB, before photobleaching, right after photobleaching (0 s), or 4 and 30 seconds after photobleaching (4 s and 30 s). Scale bar, 1 μ m. (D) The graph shows the quantification of the remaining TDP-43 fluorescence (FL) in C. Error bars indicate mean $\pm$ SD, $N = 28$ for control and 23 for PlaB-treated cells. (E) Live cell imaging of anisosome fusion in TDP-43 2KQ-Clover cells treated with 20 nM PlaB for 5 h before tracking the fusion. Representative images from Movie S1 showing fusion events indicated by arrows. Scale bar, 1 μ m. (F) Representative images of anisosome-induced (24 h) cells treated with DMSO (control) or ANS (200 nM) for 16 h. Scale bar, 1 μ m. (G) Quantification of the number of anisosomes per cell in randomly selected images of DMSO-, Cycloheximide (CHX)-, or ANS-treated cells. **** $p < 0.0001$ by one-way ANOVA; each dot represents an image with at least 20 cells counted. $n=3$ biological repeats. (H) Quantification of anisosome size in control or cells treated with CHX or ANS. ****, $p < 0.0001$ by One-way ANOVA. $n=3$ biological repeats. AU, arbitrary unit. (I) As in C except that anisosome-induced cells were treated with CHX or ANS before photobleaching. Scale bar 1 μ m. (J) Quantification of fluorescence recovery in CHX- or ANS-treated cells vs the DMSO control.

anisosomes (Figure 4F-H, Supplemental Figure S2B). FRAP analysis revealed that TDP-43 dynamics within anisosomes were similar between ANS-treated and control cells (Figure 4I, J). These results suggest that translation inhibition also stabilizes TDP-43 in liquid condensates.

Nuclear export regulates TDP-43 liquid-to-solid phase transition

Our screen also identified several nuclear transport regulators (e.g., Exportin-1/XPO1, Exportin-2/CSE1L) and nuclear pore components as anisosome regulators (Figure 3B). We focused on XPO1 because our chemical genetic screen identified two XPO1 inhibitors, KPT-276 and Verdineoxor as potent anisosome modulators (Figure 1B).

To explore the role of XPO1 in TDP-43 anisosome regulation, we induced anisosome formation in Clover-TDP-43 2KQ cells and treated them with Leptomycin B (LMB), a potent XPO1 inhibitor. LMB treatment resulted in a time- and dose-dependent reduction in anisosome number, while increasing their sizes (Figure 5A-C). The enlarged TDP-43 puncta showed the typical hollowed anisosome ring structure (Figure 5D), indicating that TDP-43 remained in the liquid state. Indeed, FRAP experiments demonstrated that LMB treatment did not significantly affect the fluorescence recovery rate of TDP-43 within bleached anisosomes (Figure 5D, E). Time-lapse confocal imaging showed that shortly after LMB treatment, preformed anisosomes began to fuse with each other (Figure 5F, Movie S2). Thus, XPO1 inhibition stabilizes TDP-43 in a liquid phase.

To corroborate the inhibitor study, we overexpressed mCherry-tagged XPO1. Surprisingly, we observed fewer and enlarged TDP-43 positive puncta in mCherry-XPO1-expressing cells (Figure 5G, panels 1, 2). At first glance, this phenotype resembled that of XPO1-inhibited cells. However, careful examination revealed several distinct features. First, despite the size increase, we did not observe the typical hollowed anisosome ring structure. Secondly, while anisosomes in mCherry-XPO1 negative cells were almost exclusively nuclear, ~30% of mCherry-XPO1-expressing cells contained cytoplasm-localized TDP-43 puncta (Figure 5G, panels 1, 2, 3, Figure 5H). Thirdly, FRAP experiments demonstrated that Clover-TDP-43 fluorescence did not recover after photobleaching, suggesting that TDP-43 was in a gel-like state transitioning to aggregates (Figure 5I). Notably, mCherry-XPO1 could be detected within TDP-43 puncta in some cells (Figure 5G, panels 3-6). These findings are consistent with a previous report (Archbold et al., 2018), demonstrating that XPO1 promotes TDP-43 nuclear egression and cytoplasmic aggregate formation. Immunostaining of endogenous XPO1 in anisosome-induced cells further confirmed this conclusion. Although antibodies could not penetrate anisosomes (Yu et al., 2021), anisosome induction depleted XPO1 from the nucleoplasm, presumably due to its sequestration into anisosomes (Figure 5J, K). Since XPO1 does not bind TDP-43 directly (Pinarbasi et al., 2018), additional factors likely facilitate XPO1-mediated TDP-43 nuclear egression under this condition. Collectively, these results suggest that the stability and dynamics of anisosomes are modulated by XPO1: reduced XPO1 activity stabilizes the liquid state and favors large TDP-43 condensates, whereas increasing XPO1 activity promotes its transition to a gel-like structure and cytoplasmic redistribution (Figure 5L). This conclusion underscores a critical regulatory link between XPO1-dependent nuclear egression and the TDP-43 liquid-to-solid phase transition.

XPO1 inhibition mitigates TDP-43 hyperphosphorylation and modest rescues neuronal defects in an anisosome-bearing organoid model

Since increased XPO1 activity is associated with more cytoplasmic, gel-like TDP-43, we tested whether reducing XPO1 activity could mitigate TDP-43 cytosolic aggregation using a recently established anisosome-bearing organoid model of TDP-43 proteinopathies. This model is derived from an induced pluripotent stem cell (iPSC) line carrying the ALS-associated mutation (K181E) (Zhang et al., 2025b), which like 2KQ disrupts the RNA binding activity of TDP-43 (Chen et al.,

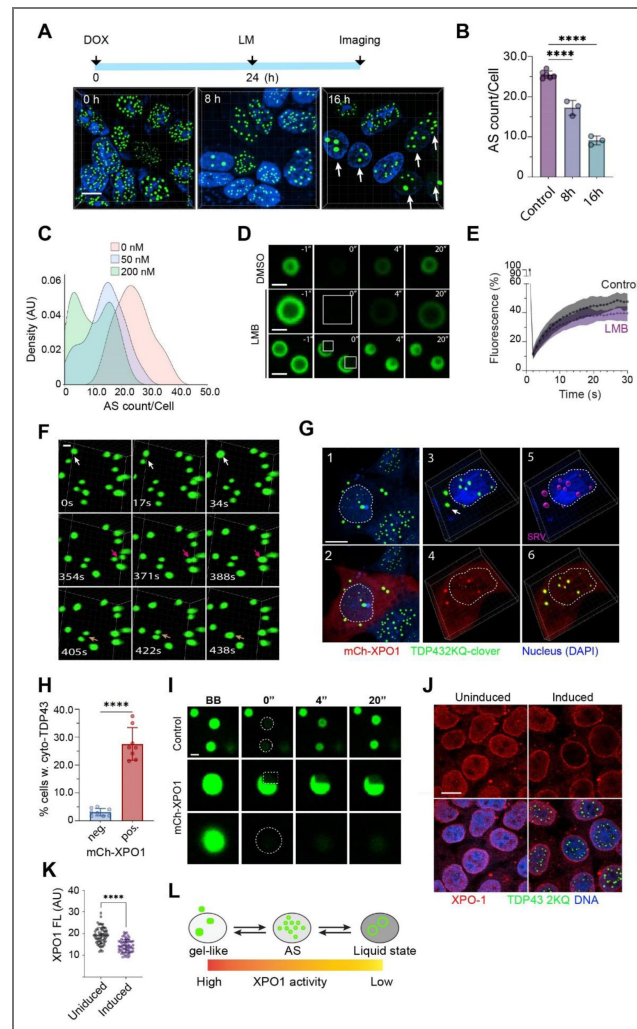


Figure 5. XPO1 regulates anisosome liquid-to-solid transition

(A) Pharmacological inhibition of XPO-1 with Leptomycin B (LMB) reduces the number of anisosome. The graph on top indicates the experimental design. Arrows show cells with enlarged anisosomes. Scale bar 10 μ m **(B)** Quantification of anisosome numbers per cell in experiments represented by A. Error bars indicate mean $\pm$ SD; **** $p < 0.0001$ by one-way ANOVA. $n=3$ biological repeats. **(C)** LMB dose dependently reduces anisosome number. Anisosome were induced in TDP-43 2KQ-Clover cells followed by treatment with LMB at the indicated concentrations for 16 h. The histogram shows the distribution of cells (50-90 cells/condition) by anisosome count. **(D)** FRAP experiments demonstrate that anisosomes remain in a liquid phase following LMB treatment (200 nM, 16 h). Anisosome-induced (24 h) cells were treated with DMSO or LMB for 16 h and then photobleached at the indicated areas. Scale bar 1 μ m. **(E)** Quantification of the FRAP experiment in D. **(F)** Time-lapse confocal microscopy detects anisosome fusion after LMB (200 nM, 5 h) treatment in TDP-43 2KQ-Clover cells. Arrows indicate fusion events. Scale bar, 1 μ m **(G)** Overexpression of mCherry-tagged XPO-1 in TDP-43 2KQ-Clover cells induces cytoplasmic TDP-43 puncta. TDP-43 2KQ-Clover cells (green) transfected with mCherry-XPO1 (red) were stained with DAPI (blue) to label nuclei (dashed lines). Cells were imaged 48 h post-transfection. Panels 1, 2 show a representative confocal section, while panels 3-6 show reconstructed 3-D views. The position and volume of anisosomes were also presented in magenta in a surface-rendered view (SRV) in panel 5. The arrow in panel 3 indicates an example of cytoplasmic TDP-43 aggregate. Scale bar, 10 μ m. **(H)** Quantification of the percentage of cells showing cytosolic TDP-43 puncta in XPO-1 positive (pos) and negative (neg) cells in randomly selected fields from 3 independent experiments. **** $p < 0.0001$ by unpaired Student's t-test. **(I)** FRAP-based confocal imaging reveals the transition of anisosomes into a gel-like state upon XPO-1 overexpression. Scale bar, 1 μ m. **(J)** Anisosome formation changes endogenous XPO-1 localization. TDP-43 2KQ-Clover before or after anisosome induction were stained with anti-XPO-1 antibodies (red) and DAPI (blue). The bleached areas were indicated by dashed lines. Scale bar, 10 μ m. **(K)** Quantification of nuclear endogenous XPO-1 levels in individual cells (indicated by dots) before or after anisosome induction. **** $p < 0.001$ by unpaired Student's t-test. $n=2$ independent repeats. **(L)** A schematic model depicting how the nuclear XPO-1 activity influences TDP-43 liquid-to-phase transition. AS, anisosome.

2019). Additionally, the K181E mutant accumulates in a liquid-condensed phase in association with HSP70, and like anisosomes, these condensates can convert to solid aggregates during cellular stress or HSP70 depletion (Gu et al., 2021).

We treated homozygous K181E organoids or wild-type (WT) controls with KPT-276 at a low dose (20 nM) for 35 days, starting on day 87 based on pilot data indicating good tolerance with prolonged treatment. Since hyperphosphorylated TDP-43 (p-TDP-43) is a major aggregate component, we stained organoids with antibodies against p-TDP-43 and total TDP-43. Confocal imaging revealed that WT and K181E organoids had similar amounts of total TDP-43, but p-TDP-43 was only significantly present in mutant organoids (Figure 6A), as reported previously (Zhang et al., 2025b). KPT-276 did not alter the total TDP-43 levels in either genotype (Figure 6A) but significantly reduced the cytosolic p-TDP-43 positive puncta in K181E organoids (Figure 6A, B).

To assess whether KPT-276 treatment also rescued the neuronal defects in TDP-43 K181E organoids, we performed bulk RNAseq using KPT-276-treated K181E and WT organoids. A comparison of treated mutant organoids to WT or untreated K181E organoids showed that KPT-276 reversed expression defects in 58 downregulated (2.90%) and 45 (1.99 %) upregulated genes (Supplemental Figure 3, Supplemental Table S4). GO pathway analysis of genes rescued from downregulation by KPT-276 highlighted neuronal pathways including neuron differentiation and brain development; for genes upregulated in K181E organoids, rescued genes are linked to G-protein coupled neurotransmitter receptor activities at the plasma membranes (Figure 6C, Supplementary Table S4). Together, these results demonstrate that XPO1 inhibition by KPT-276 modestly restores gene transcription defects in a subset of neuronal pathways in K181E mutant organoids, while robustly mitigating FTD-associated p-TDP-43 accumulation. Thus, reducing TDP-43 cytoplasmic aggregation alone may offer only limited benefit in disease treatment.

Discussion

ALS- and FTD-associated TDP-43 mutants defective in RNA binding are known to undergo self-association through their intrinsically disordered regions and RNA-binding domains, generating demixed liquid, gel, and aggregated states that remain in dynamic exchange. This protein phase behavior is shared by many RNA binding proteins and is increasingly recognized as a contributor to protein aggregation in neurodegenerative diseases (Shorter, 2019). Intriguingly, TDP-43 phase separation forms unique layered nuclear structures, with an HSP70-filled central core surrounded by a protein shell made of TDP-43. Although it is unclear how these structures are nucleated in the cell, our observations suggest that, once formed, they can grow by recruiting additional TDP-43 from the surrounding environment. Small anisosomes may also fuse with one another, although such fusion events become infrequent and difficult to detect after anisosomes reach a steady state. Our study supports the notion that TDP-43, at high concentrations, can transition from a demixing-prone, liquid state into a gel-like state en route to forming amorphous aggregates (Babinchak et al., 2019; Carey and Guo, 2022). We found that inhibition of RNA splicing, protein translation, or nuclear export favors the liquid state of TDP-43, while proteasome and HSP90 inhibition or enhancing XPO-1-mediated nuclear egression promotes the conversion of anisosomes to a gel-like state.

Because the TDP-43 2KQ mutant lacks RNA binding activity, the effects of splicing inhibition are likely mediated by other RNA-binding factors involved in splicing regulation. TDP-43 is known to interact with many splicing factors including members of the hnRNP family like hnRNP A1/A2 and PSF (Freibaum et al., 2010); Inhibition of RNA splicing may alter TDP-43's interactions with these factors, trapping it in certain complexes that favor its retention in the liquid state. Alternatively, splicing inhibition may change the overall nuclear environment, allowing large anisosomes to fuse with each other as observed in our study. This could explain why PlaB treatment reduces the number of TDP-43 anisosomes while maintaining their dynamic liquid property. Likewise, inhibition of nuclear export may influence anisosome dynamics by reshaping the nuclear environment, lowering the barrier to anisosome fusion. Future studies are required to elucidate the precise mechanisms by which these pathways regulate anisosome dynamics.

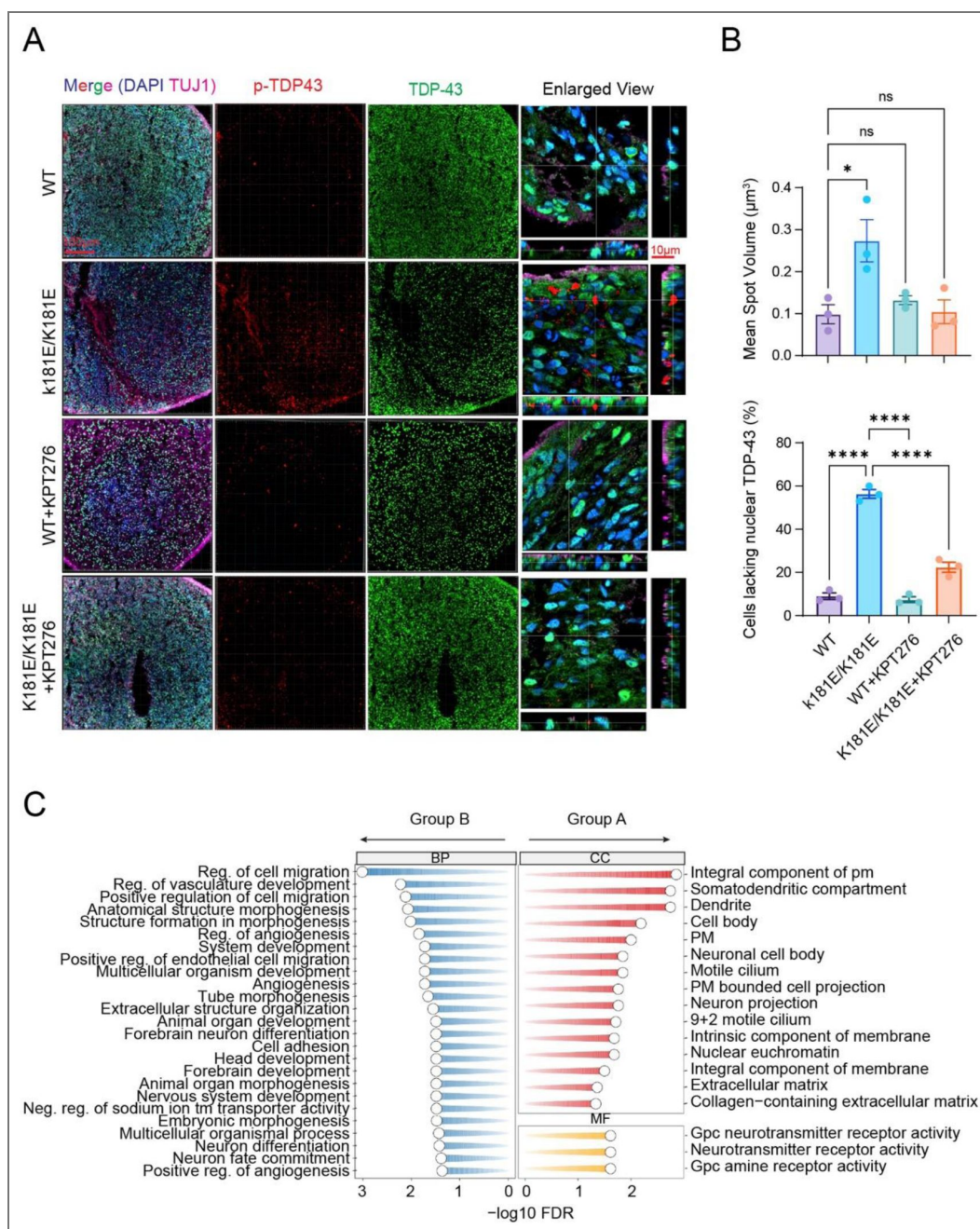


Figure 6. Inhibition of XPO-1 reduces TDP-43 hyperphosphorylation but only partially restores neuronal gene expression in K181E organoids

(A) Confocal fluorescence imaging reveals significant reduction in phosphorylated TDP-43 (p-TDP43) in K181E/K181E organoids following KPT-276 treatment. Organoids of the indicated genotypes at day 87 were treated with DMSO as a control, or with KPT-276 (20 nM) for 35 days. Organoids were fixed, sectioned, and stained with antibodies against TUJ1, a neuronal marker (Magenta), TDP-43 (green) and p-TDP-43 (red). $n = 3$ -5 organoids from two individual batches. **(B)** Quantification of p-TDP-43 Mean Spot Volume (top) and percentage of cells lacking nuclear TDP-43 (bottom) in experiments represented by **A**. Error bars indicate mean $\pm$ s.e.m., * $p < 0.05$, **** $p < 0.0001$ by one-way ANOVA, $n = 3$ organoids. **(C)** Gene Ontology (GO) analysis of genes whose expression was disrupted in K181E/K181E organoids but restored by KPT-267 as shown in Supplemental Fig. S3. BP, Biological Pathway; CC, Cellular Component; MF, Molecular Function.

Our validation of proteasome inhibition as a driver of TDP-43 droplet enlargement and hardening reinforces the idea that impaired protein clearance promotes the transition from liquid condensates to gel-like or solid aggregates. This is consistent with previous studies in ALS models, which showed that proteasome stress accelerates TDP-43 aggregation and toxicity (van Eersel et al., 2011 [↗](#)). In contrast, inhibition of deubiquitinases reduced the number of TDP-43 anisosomes, probably by promoting the proteasome-mediated clearance of TDP-43. The role of HSP90 in this process may be linked to TDP-43 folding stability, as HSP90 can bind TDP-43 to prevent promiscuous interactions (Lin et al., 2021 [↗](#)). Collectively, these observations suggest that TDP-43 condensates are maintained by a delicate balance between formation and clearance, modulated by proteolytic and chaperone activities.

While out chemical and genetic screens have identified multiple pathways influencing TDP-43 phase states, a particularly compelling aspect of our study is the discovery that the nuclear export receptor XPO1 governs TDP-43 liquid-to-solid transitions and subcellular localization. Pharmacological inhibition of XPO1 mitigates cytoplasmic p-TDP-43 accumulation, whereas XPO1 overexpression promotes the cytoplasmic accumulation of TDP-43-containing puncta. However, XPO1 may modulate TDP-43 nuclear egression indirectly, as suggested previously (Duan et al., 2022 [↗](#); Pinarbasi et al., 2018 [↗](#)). While previous studies have not detected strong interactions between XPO-1 and TDP-43 (Pinarbasi et al., 2018 [↗](#)), a weak or indirect interaction between XPO-1 and TDP-43 may still exist, especially when TDP-43 undergoes demixing. This model would explain why anisosome induction depletes endogenous XPO-1 from the nucleoplasm. Our findings also hint at a potential feedback mechanism in which TDP-43 demixing perturbs nuclear export, analogous to how cytoplasmic TDP-43 aggregates disrupt nuclear import. Given that ALS is characterized by cytoplasmic TDP-43 mis-localization and hyperphosphorylation, the XPO-1-dependent nuclear export pathway may constitute a key molecular switch linking physiological phase behavior to pathological aggregation.

Notably, in 3-D brain organoids bearing the K181E mutation, inhibition of nuclear export by KPT-276 only modestly rescues the transcriptional defects in homozygous TDP-43 K181E organoids despite an almost complete ablation of cytoplasmic phospho-TDP-43. This observation suggests that cytoplasmic TDP-43 aggregation is only a modest contributor to TDP-43 proteinopathies that lead to neurodegeneration. Major defects may still be attributed to deregulated RNA processing due to altered RNA binding (Hergesheimer et al., 2019 [↗](#); Joseph et al., 2025 [↗](#); Mehta et al., 2023 [↗](#); Zhang et al., 2025b [↗](#)). Thus, effective therapy likely requires both inhibition of cytoplasmic TDP-43 aggregation and restoration of normal RNA processing and nuclear transport. Together, our results support a model in which multiple cellular pathways including proteostasis, RNA metabolism, translation, and nuclear export—cooperatively determine the phase behavior of TDP-43. These findings have deepened our understanding of how physiological phase separation can evolve into pathogenic aggregation through cumulative failures of cellular quality control and nuclear export systems.

Methods

RNAi-based genetic screen for anisosome regulators

Genome-wide RNA interference (RNAi) screen was performed using DLD1 cells stably expressing Clover-tagged TDP-43 2KQ. Cells were transfected with siRNAs targeting 21,404 genes for 72 hours, followed by induction of anisosome formation with doxycycline (0.5 $\mu\text{g/mL}$) for 24 h. Specifically, the screen was carried out in 384-well format with 3 individual siRNAs for each gene. To prepare siRNA transfection, control siRNA (40 nmol) was mixed with 2 mL water to create a 50X (20 μM) stock and siRNA-death (20 nmol) was mixed with 1 mL of water for the same concentration. Dilute siRNA stocks to a 400 nM working concentration in water. For each well, mix 0.1 μL transfection reagent in 20 μL of cell culture medium without FBS or P/S. Add 2 μL of siRNA solution to each well and incubate with RNAiMAX for 30 minutes at room temperature. While incubating RNAiMAX, prepare trypsinized cells by ensuring thorough separation and accurate cell counting. Add 20 μL of cell suspension at a density of 0.43×10^5 cells/mL (850 cells/well) in medium containing 20% FBS and 2 x P/S to achieve a final volume of 40 μL per well. Incubate cells for 3 days before addition of

10 μ L 10 x doxycycline solution to each well (final volume 50 μ L) for anisosome induction. 24 h later, high-throughput (HT) imaging and automated phenotypic screening were conducted to identify genes that modulated anisosome formation.

Fluorescence imaging was conducted using an Opera Phenix High-Content Screening System (PerkinElmer). Whole-well images were acquired in confocal mode using a 20X water objective lens after anisosome induction for 24 h. Images were reconstructed and analyzed using the PerkinElmer Columbus server (v2.9.1). Nuclei were segmented based on Hoechst33342 staining, and anisosomes were detected via YFP signals within the nuclear region. The mean anisosome count per nucleus was calculated to assess changes in anisosome levels, while toxicity was evaluated by counting the total number of nuclei. Data were normalized to the siRNA-Neg control for each plate, and Z-scores were calculated to identify significant hits. Identified 1,533 genes whose knockdown significantly reduced the number of anisosomes per cell (Z-score > 2) were subjected to protein network analysis by STRING to identify candidate anisosome regulatory genes that were subject to a second-round screen.

A chemical genetic screen for anisosome regulators

We also performed a chemical genetic screen using LOPAC^{R1280} (Sigma) compound library on DLD1 cells expressing Clover-tagged TDP-43-2KQ. Briefly, cells were induced with doxycycline for 24 h to promote anisosome formation, followed by treatment with individual compounds for an additional 24 hours. High-throughput (HT) imaging and automated phenotypic screening were conducted to identify compounds that modulated anisosome formation.

Visualization and analysis of anisosome dynamics

To examine the impact of drug treatment on anisosome size, cells were induced to form anisosome for 24 h and then treated by different inhibitors as follows: cycloheximide (CHX, 20 μ g/mL), anisomycin (ANS, 200 nM) for 16 hours. For splicing inhibition, cells were exposed to Pladienolide-B (PlaB) at concentrations of 5 or 20 nM, or DMSO (control) for 16 hours before imaging. Cells were fixed with 4% paraformaldehyde in Phosphate Buffer Saline (PBS). Randomly selected fields were used to quantify the number of anisosomes per cell with at least two biological repeats.

To image anisosome fusion, live cell imaging was performed by treating cells with Leptomycin B (200 nM) or PlaB (20 nM) for 5 hours and then imaged by confocal 3-D sectioning for 30 minutes to capture anisosome fusion events. For Fluorescence Recovery After Photobleaching (FRAP), cells were treated with DMSO, Leptomycin B, Geldanamycin or PlaB as indicated in figure legends. Either part of an anisosome or entire anisosome was photobleached and imaged. TDP-43 fluorescence recovery after photobleaching was quantified by ImageJ and calculated to assess anisosome dynamics. At least 10 anisosomes for each condition were bleached and analyzed.

To evaluate the time course of Leptomycin B (LMB)-induced anisosome changes, DLD1 TDP-43 2KQ-Clover cells were seeded and induced with 0.5 μ g/mL doxycycline for 24 hours. Cells were treated with 10 μ M LMB and fixed at 0-, 8-, and 16-hours post-treatment. Fixed cells were stained with Hoechst to visualize nuclei. 10-20 confocal 3-D sections were obtained for each randomly selected field. Images were converted to maximum projected view by ImageJ before anisosome counting and intensity measurement. The density plot of anisosome counts per cell was generated using R, illustrating dose-dependent effects of leptomycin on anisosome formation.

To test the effect of XPO1 overexpression on TDP-43 phase regulation, DLD1 TDP-43 2KQ-Clover cells were co-transfected with mCherry-tagged XPO1 (mCh-XPO1) for 24 h before doxycycline was added at 0.5 μ g/mL to induce anisosome formation for 24 h before confocal imaging.

To study the relationship between anisosome induction and endogenous XPO1 localization, cells were seeded, induced with 0.5 μ M doxycycline for 48 hours, and fixed with paraformaldehyde in PBS. Immunostaining was performed using an XPO1 antibody (Cell Signaling, 46249S, 1:125) and Hoechst nuclear stain. Fluorescence intensity of XPO1 in both induced and uninduced cells was measured by ImageJ.

Experiments with organoids

Stem cell gene editing of the patient K181E mutation in iPSC cells was reported previously (Zhang et al., 2025b [↗](#)). Differentiation of iPSCs into forebrain neurons was performed following a previously published protocol (Fernandopulle et al., 2018 [↗](#)). Briefly, the procedure used an induction medium (IM-N2) and a neuronal culture medium (CM). IM-N2 was prepared with Knockout DMEM/F12, N2 supplement, NEAA, Gluta-MAX, Chroman I, and doxycycline. On Day 0, iPSCs were observed for confluency, dissociated with Accutase, centrifuged, and resuspended in IM-N2 with Chroman I, and then seeded into Matrigel-coated plates. Over the next four days, cells were monitored microscopically for neurite extensions while media containing doxycycline is refreshed daily. By Day 4, cells exhibit neurite growth and are ready for replating onto poly-L-ornithine (PLO)-coated dishes, prepared in advance by coating with PLO solution, incubating, washing, and drying. Replated cells were cultured in CM comprising BrainPhys medium, B27+ supplement, neurotrophic factors (GDNF, BDNF, NT-3), laminin, and doxycycline, to promote neuronal maturation.

3-D culture and organoid growth were performed as previously described (Nguyen, 2022 [↗](#)). In brief, iPSCs grown on Matrigel were dissociated into single cell suspension by Versene solution (ThermoFisher) and seeded into a 12-well Aggrewell plate (Stemcell Technologies) at 4,000 cells/well. Next day, spheroids were transferred to an ultralow attachment plate (Corning) containing phase I medium: DMEM/F12, 20% Gibco KnockOut Serum Replacement, 1X Glutamax, 1X MEM Nonessential Amino Acid, 55 μ M β -Mercaptoethanol, 1X Pen/Strep, 2 μ M Dorsomorphine, 2 μ M A83-01. After 5 days, medium was switched to phase II medium: DMEM/F12, 1X N-2 Supplement, 1X Glutamax, 1X MEM Nonessential Amino Acid solution, 1X Pen/Strep, 4 ng/mL WNT3a, 1 μ M CHIR-99021, 1 μ M SB-431542. On day 7, spheroids were embedded into Matrigel and allowed to continue growing for 7 more days. On day 14, individual spheroids were manually freed from the Matrigel and transferred to a SpinOmega bioreactor spinning at 120 RMP with phase III media: DMEM/F12, 1X N-2 Supplement, 1X B-27 Supplement, 1X Glutamax, 1X MEM Nonessential Amino Acid, 55 μ M 2-Mercaptoethanol, 1X Pen/Strep, 2.5 μ g/mL insulin. 50 days later, the medium was switched to final differentiation medium: Neurobasal medium, 1X B-27 Supplement, 1X Glutamax, 55 μ M 2-Mercaptoethanol, 1X Pen/Strep, 0.2 mM Ascorbic acid, 0.5 mM cAMP, 20 ng/mL brain-derived neurotrophic factor (BDNF), 20 ng/mL glial-derived neurotrophic factor (GDNF). 107-day-old organoids were dissociated into single cells using a 50:50 mixed of Accutase and 0.25% Trypsin with DNase I (1 mg/mL) and plated onto chambered glass slides pretreated with 1% Matrigel (Corning, Inc). Alternatively, organoids were fixed and sectioned for immunostaining (see below). For drug treatment, forebrain organoids (87 Day) were transferred to a 12-well plate on an orbital shaker (120 RPM) and treated with 20 nM KPT276 in final differentiation medium. Medium was changed every other day (Dexoregen, Inc). The organoids were harvested and frozen after 35 days of treatment for sectioning or RNAseq analysis.

Tissue preparation and immunostaining

Organoids were fixed in 4% paraformaldehyde in PBS for 1 hour at room temperature, washed with PBS, and immersed in 15% sucrose/PBS solution overnight. Subsequently, organoids were embedded in O.C.T. compound (Sakura) in a plastic mold and frozen down in an ultralow freezer. Embedded organoids were sliced onto glass slides using a Cryostat (Leica). Slides were rinsed with PBS, permeabilized with 0.5% Triton-X/PBS solution for 1 hour at room temperature and blocked using 1% donkey serum in 0.1% Tween-20/PBS solution for 1 hour. Primary antibodies, chicken anti-TUJ1 (Abcam, Ab41489, 1:1,000 dilution), rabbit anti-TDP-43 (10782-2-AP, 1/1000 dilution), mouse anti-phospho-TDP-43 (Cosmobio, CAC-TIP-PTD-M01A, 1/500 dilution), were added to the slides and the slides were incubated in a humidified chamber at 4 °C overnight. After several washes in 0.1% Tween-20/PBS solution, DAPI (Sigma) or secondary antibodies, donkey anti-rabbit, anti-mouse, and anti-chicken (Jackson ImmunoResearch), diluted 1:1,000 in blocking solution, were added to the slides. After 1 hour of incubation at room temperature, the slides were washed and mounted in antifade mounting solution (Fisher scientific).

Image acquisition and processing

Fluorescence confocal images were acquired using a Nikon CSU-W1 SoRa microscope equipped with temperature and CO₂ control enclosures and a 60x TIRF lens. Image reconstructions and analyses, including 3-D and time lapse visualizations, were performed using Imaris software or Image J. Fluorescence intensity was quantified using the open-source Fiji/Image J software. Images of randomly selected fields were split into individual channels. A consistent threshold method was applied to individual channel. The particle analysis function was used to automatically identify all anisosomes or other fluorescence structures for size and intensity measurement. Statistical analyses were conducted using Excel (for Student's t-test) or GraphPad Prism versions 8.0 and 9.0. P-values were calculated using Student's t-test in Excel or one-way ANOVA in GraphPad Prism. Curve fitting, including linear and nonlinear models, as well as IC₅₀ calculations, were also performed using GraphPad Prism versions 8.0 and 9.0.

Bulk RNA-seq library preparation and data analysis

800 ng of total RNA samples to generate the library with the NEBNext rRNA Depletion Kit v2 (Human/ mouse/rat) (NEB #E7405), NEBNext Ultra II Directional RNA Library Prep Kit for Illumina (NEB #E7760) and NEBNext® Multiplex Oligos for Illumina® (96 Unique Dual Index Primer Pairs, NEB, E6440) as described in the manufacturer's manual. Sequencing carried out in the Illumina NextSeq2000 instrument with 101×101 pair end configuration by NCI genomic core.

Adapters were trimmed using Cutadapt (v4.7), and reads were aligned to the GRCh38 genome build with STAR (v2.7.11b) using gene annotations from GENCODE v42. Gene expression was quantified with HTSeq (v2.0.7) (Putri et al., 2022 [DOI](#)), utilizing the same gene annotations. For differential gene expression analysis, all samples were processed uniformly following the standard DESeq2 (Love et al., 2014 [DOI](#)) workflow. Differential expression was defined based on an adjusted P-value threshold of <0.05 and |log₂ fold change (log₂FC) > 1. To identify gene expression rescued by exportin-1 inhibitor treatment, the fold change of expression level should be less than without inhibitor treatment, with P-value < 0.05.

Statistic and reproducibility

All statistical analyses were conducted with GraphPad Prism v10 or Excel. Statistical methods and the number of cells or brain organoid samples (N) are indicated in figure legends or shown in figures as individual data point. Biological repeats (n) are specified in figure legends. No biological repeat was excluded from the analyses. All experiments were repeated at least twice with individual data point labeled in figures unless specified. For imaging analyses, cells in randomly selected fields were analyzed. The researchers were not blinded. The iPSC cell line was derived from a ADRDs genetic risk-free clone, which was initially obtained from a male. Figures were prepared using ImageJ 1.54f, Imaris 9.9.0, Adobe Photoshop v25.12.1, and Adobe Illustrator 28.7.4.

Supplemental Figures

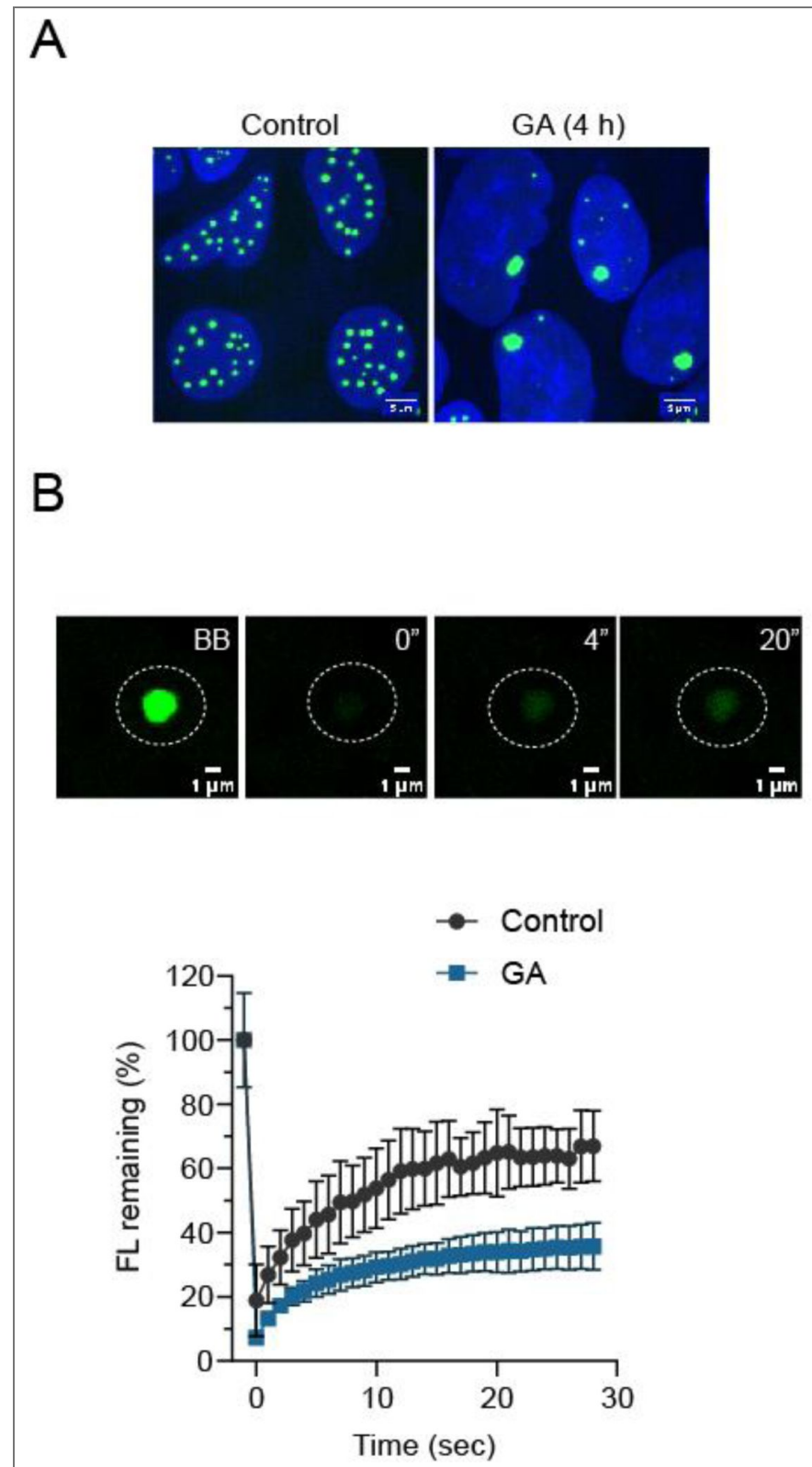


Figure S1. HSP90 inhibitor Geldanamycin converts TDP-43 anisome to a gel-like state (A) DLD1 cells were treated with doxycycline to induce anisome formation and then treated with Geldanamycin (GA) at 10 μ M for 4 h. Cells were stained with Hoechst (Blue) and imaged by a confocal microscope. Scale bar, 5 μ m. (B) A TDP-43-

bearing anisosome (dashed circle) in a GA-treated cell was photobleached and then imaged. Scale bar, 1 μm . The graph shows the quantification. Error bars, s.e.m. ($N > 5$ anisosomes). FL, fluorescence.

Figure S2. The impact of splicing inhibitor and translation inhibitor on anisosomes

(A) A violin plot showing the relative size distribution of anisosomes (AS) in DLD1 cells treated with Pladienolide-B at the indicated concentration for 16 h. ****, $p < 0.0001$ by one-way ANOVA. $n = 3$ biological repeats. (B) DLD1 cells were treated with doxycycline to induce anisome formation and then treated with Cycloheximide (CHX) at $20 \mu\text{g/mL}$ or DMSO as a control for 16 h. Cells were stained with Hoechst (Blue) and imaged by a confocal microscope. Scale bar, $5 \mu\text{m}$.

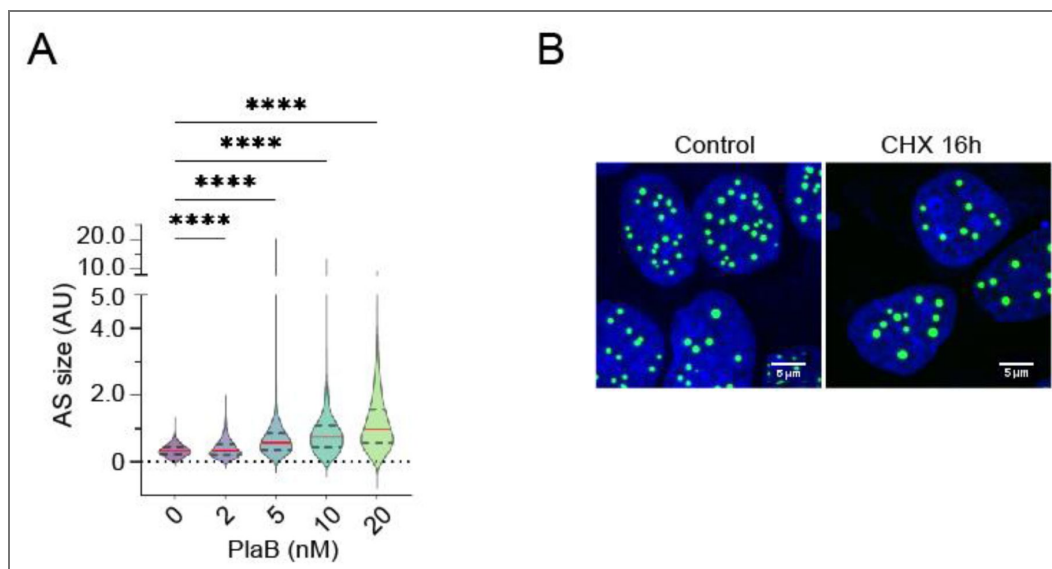
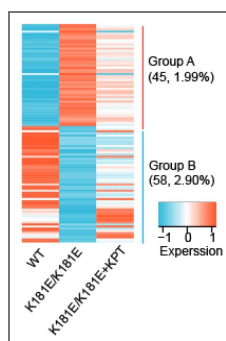


Figure S3. KPT-276 modestly reverses gene expression defects in K181E mutant organoids

Heatmap showing genes whose expression was altered in K181E/K181E organoids and restored after KPT-276 treatment.



Data availability

Raw and processed bulk RNA-seq data can be accessed from Sequence Read Archive (SRA), NCBI with Bioproject ID: PRJNA1195353, <http://www.ncbi.nlm.nih.gov/bioproject/1195353> .

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Additional information

Author contributions

YY conceived and designed the study. YY, KC and WZ supervised the study. QZ and KC conducted the chemical and siRNA screens. NC performed post-screen validation and characterization of anisosomes under different conditions. J Z constructed the iPSC knock-in lines and QZ conducted the organoid study. QZ, NC and YY analyzed the data. YY wrote the manuscript. All authors read, edited, and approved the manuscript.

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Additional files

Movie S1  PlaB treatment induces anisosome fusion.

Movie S2  Blocking XPO-1-mediated nuclear export induces anisosome fusion.

Table S1  A chemical genetic screen identified compounds that reduce anisosome numbers.

Table S2  A genome-wide siRNA screen identified potential anisosome regulators.

Table S3  Top 110 genes confirmed from a secondary screen as modulators of anisosome dynamics.

Table S4  Differentially expressed genes in K181E organoids rescued by KPT-276 (KPT).

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Peer reviews

Reviewer #1 (Public review):

In this paper, the authors use a doxycycline-inducible DLD1 cell line expressing a Clover-tagged RNA-binding-defective TDP-43 2KQ mutant that forms nuclear "anisosomes" (TDP-43 shell with HSP70 core) to carry out a small-molecule screen using the LOPAC 1280 library to identify compounds that reduce anisosome number or shift their morphology and dynamics. They also conducted a genome-wide siRNA screen to identify genetic modifiers of anisosome formation and dynamics. From these screens, the authors identify pathways in RNA splicing, translation, proteostasis (proteasome and HSP90), and nuclear transport, including XPO1. They then focus on XPO1 as their primary hit. Pharmacological inhibition of XPO1 using KPT-276, Verdinexor, and Leptomycin B reduces anisosome number while enlarging remaining condensates, which retain liquid-like behavior by FRAP and fusion assays. XPO1 overexpression causes fewer, enlarged TDP-43 puncta, including cytoplasmic puncta, with little or no FRAP recovery, interpreted as gel or solid-like aggregates. Anisosome induction reduces detectable nucleoplasmic XPO1 staining. Finally, the authors examine a homozygous TDP-43 K181E iPSC-derived forebrain organoid model, showing increased cytosolic pTDP-43 in K181E/K181E organoids compared to wild-type controls. Chronic low-dose KPT-276 reduces cytoplasmic pTDP-43 without changing total TDP-43 levels. Bulk RNA-seq shows only a modest fraction of dysregulated genes in K181E/K181E organoids are rescued by KPT-276. They conclude that nuclear export, via XPO1, is a key regulator of TDP-43 liquid-to-solid phase transitions and that cytoplasmic aggregation per se may contribute only modestly to TDP-43 proteinopathy, with RNA-processing defects being dominant.

The study presents well-executed chemical and genome-wide siRNA screens in a DLD1 TDP-43 2KQ anisosome model and follows up on nuclear transport, particularly XPO1, as a modulator of TDP-43 phase behavior and cytoplasmic aggregation. The screens are impressive in scale, and the microscopy and fluorescence recovery after photobleaching (FRAP) work is technically strong. However, the central mechanistic and disease-relevance claims are not yet sufficiently supported. There are major concerns about the heavy reliance on non-physiological, RNA-binding-defective, and acetylation-mimetic TDP-43 (2KQ) and a homozygous TDP-43 K181E organoid model. An underdeveloped and partly contradictory mechanistic link exists between XPO1 and TDP-43 phase transitions in the context of prior work showing TDP-43 is not a canonical XPO1 cargo. The paper also appears to overinterpret organoid data to conclude that cytoplasmic TDP-43 aggregation plays only a minor role in

pathology, based largely on pTDP-43 antibody staining with limited sensitivity and relatively modest rescue readouts. A deeper mechanistic analysis and additional, more physiological validation are needed for this to reach the level of rigor and impact implied by the title and abstract. The work feels screen-rich but conceptually underdeveloped, with key claims outpacing the data. A major revision with substantial new data and tempering of conclusions is warranted. I outline several problematic areas below:

(1) The central mechanistic discoveries are derived almost entirely from a DLD1 colon cancer cell line overexpressing an RNA-binding-defective, acetylation-mimetic TDP-43 2KQ mutant and homozygous TDP-43 K181E iPSC-derived organoids. Both systems are far from physiological. The 2KQ mutation is a synthetic double lysine-to-glutamine mutant originally designed to mimic acetylation and disrupt RNA binding. In this study, essentially all cell-based mechanistic data on phase behavior, screens, and XPO1 effects rely on 2KQ. Yet there is no quantification of how much endogenous TDP-43 is acetylated in degenerating human neurons, nor whether a 2KQ-like acetylation state is ever achieved *in vivo*. It is not established that the phase behavior of 2KQ recapitulates the physiological or pathological phase behavior of wild-type TDP-43 or genuine disease-linked mutants, which may retain partial RNA binding and different post-translational modification patterns. As a result, it is difficult to know whether the modifiers identified here regulate a highly artificial 2KQ condensate or physiologically relevant TDP-43 condensates. To address this concern, the paper would benefit from quantifying endogenous TDP-43 acetylation at the relevant lysines in control and ALS/FTD patient tissue or more disease-proximal models such as heterozygous TARDBP mutant iPSC neurons, which would justify the focus on an acetyl-mimetic mutant. Key phenomena, including XPO1 dependence of phase behavior, effects of proteasome and HSP90 inhibition, and effects of splicing and translation inhibitors, should be tested for wild-type TDP-43 expressed at near-physiological levels and for one or more bona fide ALS/FTD-linked TARDBP mutants that are not acetyl mimetics. At a minimum, the authors should show that endogenous TDP-43 in neuronally differentiated cells exhibits qualitatively similar responses to XPO1 modulation, rather than exclusively relying on DLD1 2KQ overexpression.

(2) The organoid model is based on a homozygous K181E knock-in line. However, in patients, TARDBP mutations are overwhelmingly heterozygous. Homozygosity is thus a severe, arguably non-physiological sensitized background that may exaggerate nuclear RNA mis-splicing and phase defects and alter the relative contribution of cytoplasmic aggregation versus nuclear loss-of-function. In addition, it is not fully clear from this manuscript whether the structures in K181E organoids are bona fide anisosomes as defined in Yu et al. 2021, characterized by HSP70-enriched central liquid cores with TDP-43 shells and similar FRAP and fusion behavior to anisosomes in the DLD1 model. At present, the organoid section is framed as validation of "anisosome-bearing organoids," but the figures in this manuscript mainly show pTDP-43 puncta and total TDP-43 immunostaining, without detailed structural or biophysical characterization. The authors should explicitly compare heterozygous K181E/+ organoids or another heterozygous TARDBP mutant line with homozygous K181E/K181E organoids to assess whether XPO1 inhibition has similar effects in a genotype that more closely resembles patient genetics. They should provide direct evidence that the K181E condensates in organoids are anisosomes through HSP70 core immunostaining, three-dimensional reconstruction, and FRAP measurements, and clarify whether KPT-276 is acting on anisosome-like structures or more generic cytoplasmic aggregates or puncta. Without this, the leap from a DLD1 2KQ cancer cell model to human ALS/FTD-relevant neurons is not convincingly supported.

(3) The title and framing assert that "nuclear export governs TDP-43 phase transitions." However, prior studies such as Pinarbasi et al. 2018 and Duan et al. 2022 indicate that TDP-43 is not a canonical XPO1 cargo and that its export is largely passive, with active nuclear import being the dominant determinant of nuclear localization. The authors cite these studies but still position XPO1 as a central, quasi-direct regulator. The data presented are largely

correlative or based on pharmacologic manipulation and overexpression in an overexpression mutant background, with no direct evidence that XPO1 engages TDP-43 in a specific, regulated manner. Even if XPO1 does not engage WT TDP-43, it could still engage the 2KQ variant, which needs to be tested.

(4) The XPO1 perturbations yield somewhat confusing phenotypes. XPO1 inhibition using Leptomycin B, KPT-276, and Verdinexor reduces anisosome number and enlarges remaining anisosomes, which remain liquid-like by FRAP recovery and fusion assays and stay nuclear. XPO1 overexpression causes fewer, enlarged puncta, but these are FRAP-impaired (gel-like) and redistribute to the cytoplasm. Thus, both decreased and increased XPO1 activity reduce anisosome number and enlarge puncta, but with opposite phase behaviors and subcellular localizations. The model presented in Figure 5L is relatively qualitative and does not resolve these issues. Moreover, XPO1 inhibition globally impairs nuclear export of many cargos and profoundly alters the nuclear environment, transcription, RNA processing, and chromatin. It is therefore difficult to conclude that the observed effects are specific to TDP-43 phase regulation as opposed to secondary consequences of broad nuclear export blockade.

(5) The authors show that anisosome induction depletes nucleoplasmic XPO1 signal and that mCherry-XPO1 can be seen in some TDP-43 puncta. However, antibody penetration into anisosomes is limited, so XPO1 depletion from nucleoplasm could reflect sequestration in the anisosome shell or core, but this is not demonstrated. There is no demonstration of physical interaction, even indirect interaction, between XPO1 and TDP-43 or a defined adaptor, nor identification of a specific mutant of XPO1 that selectively disrupts this putative interaction while preserving other functions. The known TDP-43 NES has been shown to be weak and not a functional XPO1-dependent NES in multiple studies. If XPO1 is acting through an adaptor that recognizes 2KQ or K181E specifically, that by itself would bring into question the generality of the mechanism for wild-type TDP-43.

(6) To support a mechanistic claim that nuclear export governs TDP-43 phase transitions, more targeted evidence is needed. The authors should test whether siRNA knockdown or CRISPR interference of XPO1 in the DLD1 2KQ model reproduces the effects seen with Leptomycin B and KPT-276, including FRAP and fusion phenotypes, and verify on-target effects by rescue with an siRNA-resistant XPO1 construct. They should demonstrate that canonical XPO1 cargos behave as expected under the inhibitor conditions used, as a positive control, and that the concentrations used are not grossly toxic. They should attempt to identify or at least constrain candidate adaptors that might enable XPO1-dependent export of TDP-43 through proteomic analysis of XPO1 co-purifying with 2KQ condensates or loss-of-function studies of candidate adaptors from the siRNA screen. Finally, they should test whether a TDP-43 mutant that cannot bind the proposed adaptor still responds to XPO1 manipulation.

(7) Even with these data, what is currently shown is that global modulation of nuclear export capacity can alter the phase behavior and localization of a highly overexpressed RNA-binding-defective TDP-43 mutant and of K181E in organoids. This is important, but it is weaker than asserting that XPO1 directly governs TDP-43 phase transitions in physiological contexts. The title, abstract, and Discussion should be tempered to reflect that nuclear export is one of several pathways, alongside RNA splicing, translation, and proteostasis, that influence TDP-43 phase states in this model, and that the specific mechanism and cargo relationship between XPO1 and TDP-43 remain unresolved and may be indirect.

(8) The authors conclude that cytoplasmic TDP-43 aggregation plays only a modest role in TDP-43 proteinopathies because in homozygous K181E organoids, chronic KPT-276 treatment almost abolishes cytoplasmic pTDP-43 puncta, yet bulk RNA-seq shows only a relatively small fraction of dysregulated genes are rescued. There are several issues with this inference. Relying primarily on pTDP-43 antibody staining to define cytoplasmic TDP-43 aggregation is limiting. pTDP-43 antibodies label only phosphorylated species and may miss non-

phosphorylated, oligomeric, or amorphous TDP-43 species that could still be toxic. Different pTDP-43 antibodies vary in epitope accessibility depending on aggregate conformation and subcellular location. More sensitive approaches, such as high-affinity TDP-43 RNA aptamer probes developed by Gregory and colleagues, biochemical fractionation for SDS-insoluble and urea-soluble TDP-43, and filter-trap assays, would provide a more quantitative assessment of cytoplasmic aggregation and its reduction by KPT-276. Without these, it is not safe to assume that cytoplasmic aggregation has been eliminated, as opposed to one antigenic subclass.

(9) The treatment window, spanning from day 87 to 122 with 20 nanomolar KPT-276, may be too late or too mild to reverse entrenched nuclear RNA-processing defects, even if cytoplasmic inclusions are cleared. Once widespread cryptic exon inclusion and alternative polyadenylation misregulation are established, many downstream changes may become self-sustaining or only partially reversible. Moreover, XPO1 inhibition will massively rewire nucleocytoplasmic transport of many transcription factors, splicing factors, and RNA-binding proteins. Thus, the lack of full transcriptomic rescue cannot be cleanly interpreted as evidence that cytoplasmic aggregates are only modest contributors. It may instead reflect that nuclear dysfunction is primary and XPO1 inhibition does not correct, and may even exacerbate, certain nuclear defects.

(10) To support a causal statement about the modest contribution of cytoplasmic aggregates, one would want more direct measures of neuronal health and function, such as cell death, neurite complexity, synaptic markers, and electrophysiology before and after KPT-276, not only transcriptomics. A way to selectively reduce cytoplasmic aggregation without globally inhibiting nuclear export would allow comparison of outcomes.

(11) Given these caveats, the concluding statements that cytoplasmic TDP-43 aggregation is only a modest contributor should be substantially softened. A more defensible interpretation is that in this homozygous K181E organoid model, chronic global XPO1 inhibition reduces pTDP-43-positive cytoplasmic puncta but only partially normalizes the steady-state transcriptome, suggesting that persistent nuclear RNA-processing defects and other pathways continue to drive pathology.

(12) The screens are a major strength but need more rigorous validation for key hits, especially nuclear transport factors. For the siRNA screen, hits are filtered by anisome number per nucleus, but there is no direct demonstration in the main text that XPO1 or CSE1L knockdown is efficient at the messenger RNA or protein level. For the highlighted genes, Western blot or quantitative polymerase chain reaction validation and phenotypic rescue would strengthen confidence. For small-molecule hits, it is not systematically shown that anisome modulation is independent of changes in total TDP-43 2KQ expression or gross toxicity. Translation inhibitors are tested for this, but for many other hits, including proteasome, HSP90, and kinase inhibitors, expression and general nuclear structure should be monitored. Given the reliance on anisome count as a readout, secondary screens that specifically distinguish changes in TDP-43 expression levels, changes in nuclear morphology or cell cycle, and specific changes in anisome phase behavior, including FRAP and fusion for top hits, would greatly increase interpretability.

(13) The classification of condensates as liquid versus gel-like or solid is based almost entirely on FRAP recovery or lack thereof. While FRAP is appropriate, interpretations could be made more robust by including half-region-of-interest bleach controls and assessing mobile fractions and recovery kinetics more quantitatively across conditions. Complementing FRAP with other phase-behavior assays such as sensitivity to 1,6-hexanediol, shape relaxation after deformation, and coarsening behavior over longer timescales would strengthen the analysis. At present, some assignments, such as that XPO1 overexpression drives a gel-like transition, are reasonable but somewhat qualitative.

(14) For the Leptomycin B and KPT-276 experiments in cells and organoids, it would be important to confirm that canonical XPO1 cargo proteins accumulate in the nucleus and that the concentrations used are within a range that is not overtly toxic over the experimental timeframe. Assessing nuclear morphology, chromatin condensation, and general transcriptional activity through global RNA synthesis or key reporter genes would ensure that observed effects are not secondary to severe global nuclear export collapse.

(15) In the organoid section, it is not clear how many independent iPSC clones and organoid batches were used per condition, nor whether batch effects were assessed in the bulk RNA-seq analysis. This should be fully specified and ideally controlled with isogenic wild-type and K181E clones. For transcriptional rescue, it is important to know whether the changes in wild-type organoids treated with KPT-276 are negligible. A direct wild-type comparison with or without KPT-276 is important to disentangle general drug effects from K181E-specific rescue. More detailed quantification of total TDP-43 and pTDP-43 in both nuclear and cytoplasmic fractions, including biochemical fractionation if possible, would strengthen the assertion that KPT-276 specifically reduces cytosolic pTDP-43 aggregates while sparing nuclear TDP-43.

(16) Beyond the core issues above, several additions could greatly enhance the impact. The manuscript currently emphasizes XPO1, but the genetic and chemical data clearly implicate RNA splicing, translation, and proteostasis as equally strong or stronger regulators of TDP-43 phase states. A more integrated model that explains how these pathways intersect, for example, how splicing factor availability, ribosome loading, and proteasome capacity co-govern anisosome nucleation, growth, and hardening, would be valuable.

(17) A key unresolved question is whether XPO1 is acting directly on TDP-43, or instead primarily regulates anisosomes by exporting other factors that more proximally control TDP-43 phase behavior. Given that TDP-43 is not a canonical XPO1 cargo and prior work indicates that its nuclear export is largely passive, it seems at least as plausible that XPO1 inhibition alters the nuclear concentration or localization of splicing factors, RNA-binding proteins, chaperones, or other modifiers identified in the screens, and that changes in these proteins secondarily reshape anisosome dynamics. In other words, XPO1 may be exporting a more direct regulator of anisome formation and hardening, rather than exporting TDP-43 itself in a specific, regulated way. The current data do not distinguish between these possibilities. Systematic identification of XPO1-dependent cargos that colocalize with or biochemically associate with anisosomes, combined with targeted perturbation of their nuclear export, would be needed to determine whether the relevant XPO1 substrate in this system is actually TDP-43 or an upstream modulator of its phase behavior.

(18) Testing whether identified modifiers converge on nuclear TDP-43 concentration would be informative. Since phase separation is concentration-dependent, measuring nuclear versus cytoplasmic TDP-43 levels across key perturbations, including splicing inhibition, translation inhibition, proteasome inhibition, HSP90 inhibition, and XPO1 modulation, would help determine whether modifiers mainly work by changing nuclear TDP-43 concentration or by altering interaction networks and the material properties of condensates.

(19) Examining other ALS-relevant RNA-binding proteins would be valuable. Given the role of XPO1 and other hits, it would be informative to briefly test whether similar principles apply to FUS, hnRNPA1, or other ALS-relevant RNA-binding proteins in the same cellular context, to argue for generality versus TDP-43-specific idiosyncrasies of the 2KQ system.

(20) The Introduction sometimes implies that anisosomes are common and well-established intermediates en route to pathology. It would be helpful to more clearly state that, to date, anisosomes are primarily observed in overexpression and mutant systems and have not yet been unequivocally demonstrated in human patient tissue. The link between PDGFR β , PAK4, GSK-3 β , and YAP and TDP-43 phase dynamics is intriguing but only briefly mentioned. The authors should either expand on this or tone down the emphasis in the Results section.

(21) In the organoid methods, the authors should consider clarifying whether doxycycline is continuously used, which might alter TDP-43 expression and nuclear transport in a non-negligible way.

(22) For statistical methods, it would be beneficial to indicate whether multiple-comparison corrections were applied for the many FRAP, anisosome count, and size comparisons beyond DESeq2 internal corrections for RNA-seq.

(23) Some figure legends could more clearly indicate whether the images shown are single z-planes or maximum intensity projections and how the thresholding for anisosome detection was performed.

(24) In its current form, the manuscript contains an impressive set of screens and some nicely executed imaging of TDP-43 condensates, highlighting nuclear export among other pathways as a modulator of TDP-43 phase behavior. However, the physiological relevance is undercut by heavy reliance on an acetylation-mimetic, RNA-binding-defective TDP-43 mutant and a homozygous K181E organoid model. The mechanistic link between XPO1 and TDP-43 remains largely inferential and partly at odds with prior work. The conclusion that cytoplasmic TDP-43 aggregation is only a modest contributor to disease is not firmly supported by the available data.

(25) With substantial additional mechanistic work, particularly around XPO1, rigorous validation in more physiological TDP-43 contexts, more sensitive detection of cytoplasmic TDP-43 aggregates, and a tempering of the central claims, this study could make a meaningful contribution to understanding how nucleocytoplasmic transport and other cellular pathways influence TDP-43 phase transitions and aggregation. The work should be reframed as an important screening study that identifies nuclear export as one among several cellular processes that modulate TDP-43 phase behavior in a model system, rather than as a definitive demonstration that nuclear export governs pathological TDP-43 aggregation in disease.

<https://doi.org/10.7554/eLife.110172.1.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript addresses an important and timely question in TDP-43 biology by systematically identifying regulators of TDP-43 anisosome formation, with a particular focus on nuclear export via XPO1. Using a combination of unbiased chemical screening, genetic perturbation, and advanced imaging approaches, the authors propose that inhibition of nuclear export modulates the abundance and biophysical properties of TDP-43 anisosomes. The study is conceptually innovative and has potential relevance for neurodegenerative diseases characterized by TDP-43 pathology. However, significant concerns regarding experimental controls, reporting transparency, and model translatability currently limit the strength of the conclusions and the interpretability of several key findings.

Strengths:

(1) The study employs an unbiased, hypothesis-free compound screen to identify regulators of TDP-43 anisosome formation, which is a major strength and reduces confirmation bias.

(2) The authors combine chemical and genetic screening approaches, providing orthogonal validation of key pathways and increasing confidence in the biological relevance of top hits.

(3) The focus on biophysical properties of TDP-43 assemblies, assessed through imaging and FRAP, moves beyond simple presence/absence of aggregates and provides mechanistic insight into the biophysical states of TDP-43.

(4) The use of multiple experimental modalities, including live-cell imaging, FRAP, pharmacological perturbation, and transcriptomic analysis, reflects a technically sophisticated and ambitious study design.

(5) The authors attempt to extend findings beyond immortalized cancer cell lines by incorporating organoid models, demonstrating awareness of disease relevance and translational importance.

Overall, the manuscript is clearly written and logically structured, making complex experimental workflows accessible and the central hypotheses easy to follow.

Weaknesses:

Despite its strengths, the manuscript has several major limitations that affect data interpretation and confidence in the conclusions.

(1) Lack of appropriate controls for overexpression experiments:

A central concern is the absence of proper controls for TDP-43 and XPO1 overexpression. Prior studies (including those cited by the authors, Archbold et al.2018) show that overexpression of WT TDP-43 alone is toxic to neurons. Thus, the experimental system itself may induce anisome formation independently of the mechanisms under study. Similarly, XPO1 overexpression lacks a suitable control (e.g., mCherry alone or mCherry fused to a protein known to be independent of TDP-43). The near-complete colocalization of XPO1 with TDP-43 anisomes upon overexpression raises the possibility that these structures reflect non-physiological protein accumulation rather than regulated assemblies.

1. Insufficient experimental and analytical transparency:

The manuscript frequently lacks clear reporting of experimental details. In multiple figures, the stated number of independent experiments does not match the number of data points shown, making it difficult to assess statistical validity. Concentrations used in the compound screen are not clearly defined, nor is it stated whether multiple concentrations were tested. It is unclear how many wells, cells, or independent cultures were analyzed. The criteria used to reduce 1,533 screening hits to 211 candidates via STRING analysis are not explained. Knockdown and overexpression efficiencies are not reported.

(3) RNA-seq concerns:

The RNA-seq experiments are particularly problematic. The number of biological replicates per condition is not stated, and heatmaps suggest that only one sample per group may have been used, which would preclude statistical analysis. No baseline comparison between WT and mutant TDP-43 is shown. Given that TDP-43 is an RNA-binding protein, splicing analyses would be far more informative than gene expression alone, yet no splicing data are presented. Moreover, nuclear retention of TDP-43 does not preclude nuclear aggregation, which may still impair its splicing function.

(4) Limited translatability to neuronal biology:

All anisome analyses are performed in a cancer cell line, raising concerns about relevance to post-mitotic neurons. While organoids are used as a secondary model, the assays performed do not overlap with those used in cancer cells, making it difficult to assess whether anisome-related mechanisms are conserved. Neuronal toxicity, a critical outcome given known TDP-43 biology, is not assessed. Prior work has shown that WT TDP-43 overexpression alone is toxic to neurons, yet this is not addressed.

(5) Conceptual and interpretational gaps:

The authors quantify anisosome number but also report conditions in which anisosome number decreases while size increases. The biological interpretation of larger anisosomes is not discussed, and whether this reflects improvement or worsening of pathology is unclear. Compounds targeting the same mechanism (e.g., nuclear export inhibition) are inconsistently used across experiments (KPT compounds, verdinexor, leptomycin B), raising concerns about reproducibility. In organoids, the experimental paradigm shifts to long-term treatment (35 days vs. 16 hours), further complicating interpretation.

(6) Overinterpretation of rescue effects:

Although the authors state that they aim to test whether nuclear export inhibition rescues neuronal defects, no functional neuronal readouts are provided (e.g., viability, morphology, axon outgrowth, or electrophysiological measures). RNA-seq alone is insufficient to support claims of rescue.

(7) Finally, the model does not appear to exhibit cytosolic TDP-43 aggregation at baseline. It remains unclear whether longer induction would produce cytosolic gel-like assemblies and whether these would be prevented by nuclear export inhibition. Long-term data are shown only in organoids, yet anisosome formation is not assessed there.

<https://doi.org/10.7554/eLife.110172.1.sa2>

Reviewer #3 (Public review):

Summary:

TDP-43 proteinopathy is broadly found in neurodegenerative diseases. This manuscript investigates how nuclear export influences the biophysical properties of TDP-43. The authors use a combination of chemical screening and genome-wide siRNA screening to identify pathways that modulate TDP-43 liquid-to-solid transitions. Overall, the study employs a broad array of approaches and addresses an important question in TDP-43 pathobiology. The identification of nuclear export as a central regulator is compelling and conceptually aligns with the emerging view that TDP-43 nucleocytoplasmic trafficking is a major defect in neurodegeneration.

Strengths:

This work integrates chemical and genetic screening to identify novel modifiers. The candidates were validated in both reporter cell lines and iPS-differentiated organoids. The findings support the nucleocytoplasmic transport is important for the biophysical properties of TDP-43.

Weaknesses:

The mechanisms underlying the connection between nuclear export and phase transition need further clarification. Broader consequences of XPO1 inhibition are not addressed.

<https://doi.org/10.7554/eLife.110172.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

In this paper, the authors use a doxycycline-inducible DLD1 cell line expressing a Clover-tagged RNA-binding-defective TDP-43 2KQ mutant that forms nuclear "anisosomes" (TDP-

43 shell with HSP70 core) to carry out a small-molecule screen using the LOPAC 1280 library to identify compounds that reduce anisosome number or shift their morphology and dynamics. They also conducted a genome-wide siRNA screen to identify genetic modifiers of anisosome formation and dynamics. From these screens, the authors identify pathways in RNA splicing, translation, proteostasis (proteasome and HSP90), and nuclear transport, including XPO1. They then focus on XPO1 as their primary hit. Pharmacological inhibition of XPO1 using KPT-276, Verdinexor, and Leptomycin B reduces anisosome number while enlarging remaining condensates, which retain liquid-like behavior by FRAP and fusion assays. XPO1 overexpression causes fewer, enlarged TDP-43 puncta, including cytoplasmic puncta, with little or no FRAP recovery, interpreted as gel or solid-like aggregates. Anisosome induction reduces detectable nucleoplasmic XPO1 staining. Finally, the authors examine a homozygous TDP-43 K181E iPSC-derived forebrain organoid model, showing increased cytosolic pTDP-43 in K181E/K181E organoids compared to wild-type controls. Chronic low-dose KPT-276 reduces cytoplasmic pTDP-43 without changing total TDP-43 levels. Bulk RNA-seq shows only a modest fraction of dysregulated genes in K181E/K181E organoids are rescued by KPT-276. They conclude that nuclear export, via XPO1, is a key regulator of TDP-43 liquid-to-solid phase transitions and that cytoplasmic aggregation per se may contribute only modestly to TDP-43 proteinopathy, with RNA-processing defects being dominant.

We thank the reviewer for carefully summarizing our study.

The study presents well-executed chemical and genome-wide siRNA screens in a DLD1 TDP-43 2KQ anisosome model and follows up on nuclear transport, particularly XPO1, as a modulator of TDP-43 phase behavior and cytoplasmic aggregation. The screens are impressive in scale, and the microscopy and fluorescence recovery after photobleaching (FRAP) work is technically strong. However, the central mechanistic and disease-relevance claims are not yet sufficiently supported. There are major concerns about the heavy reliance on non-physiological, RNA-binding-defective, and acetylation-mimetic TDP-43 (2KQ) and a homozygous TDP-43 K181E organoid model. An underdeveloped and partly contradictory mechanistic link exists between XPO1 and TDP-43 phase transitions in the context of prior work showing TDP-43 is not a canonical XPO1 cargo. The paper also appears to overinterpret organoid data to conclude that cytoplasmic TDP-43 aggregation plays only a minor role in pathology, based largely on pTDP-43 antibody staining with limited sensitivity and relatively modest rescue readouts. A deeper mechanistic analysis and additional, more physiological validation are needed for this to reach the level of rigor and impact implied by the title and abstract. The work feels screen-rich but conceptually underdeveloped, with key claims outpacing the data. A major revision with substantial new data and tempering of conclusions is warranted. I outline several problematic areas below:

(1) The central mechanistic discoveries are derived almost entirely from a DLD1 colon cancer cell line overexpressing an RNA-binding-defective, acetylation-mimetic TDP-43 2KQ mutant and homozygous TDP-43 K181E iPSC-derived organoids. Both systems are far from physiological. The 2KQ mutation is a synthetic double lysine-to-glutamine mutant originally designed to mimic acetylation and disrupt RNA binding. In this study, essentially all cell-based mechanistic data on phase behavior, screens, and XPO1 effects rely on 2KQ. Yet there is no quantification of how much endogenous TDP-43 is acetylated in degenerating human neurons, nor whether a 2KQ-like acetylation state is ever achieved in vivo. It is not established that the phase behavior of 2KQ recapitulates the physiological or pathological phase behavior of wild-type TDP-43 or genuine disease-linked mutants, which may retain partial RNA binding and different post-translational modification patterns. As a result, it is difficult to know whether the modifiers identified here regulate a highly artificial 2KQ condensate or physiologically relevant TDP-43 condensates. To address this concern, the paper would benefit from quantifying

endogenous TDP-43 acetylation at the relevant lysines in control and ALS/FTD patient tissue or more disease-proximal models such as heterozygous TARDBP mutant iPSC neurons, which would justify the focus on an acetyl-mimetic mutant. Key phenomena, including XPO1 dependence of phase behavior, effects of proteasome and HSP90 inhibition, and effects of splicing and translation inhibitors, should be tested for wild-type TDP-43 expressed at near-physiological levels and for one or more bona fide ALS/FTD-linked TARDBP mutants that are not acetyl mimetics. At a minimum, the authors should show that endogenous TDP-43 in neuronally differentiated cells exhibits qualitatively similar responses to XPO1 modulation, rather than exclusively relying on DLD1 2KQ overexpression.

Acetylation of endogenous TDP-43 was reported by several studies. Although it occurs at low levels under normal conditions, TDP-43 acetylation is upregulated under stress conditions (e.g. oxidative stress and proteotoxic stress) (PMID: 25556531; PMID: 28724966). Importantly, Cohen et al. reported the identification of acetylated TDP-43 in ALS patient spinal cord (PMID: 25556531), while Yu et al. showed that endogenous wildtype TDP-43 undergoes demixing when neurons were treated with either a deacetylase inhibitor or proteasome inhibitors (PMID: 33335017). These studies also show that acetylated TDP-43 is defective in RNA binding and more prone to aggregation. Furthermore, ectopic expression of acetylated TDP-43 mimetics in cells and mice induces cellular defects similar to those observed in disease models (PMID: 28724966). Thus, our findings, based on previously established TDP-43 mimetics, should provide valuable information regarding the regulation of TDP-43 phase behavior. We agree with the reviewers that the model used in this study has its limitations, and we will be happy to revise the manuscript to tone down some conclusions, and include more background information to justify the use of TDP-43 acetylation mimetics.

(2) The organoid model is based on a homozygous K181E knock-in line. However, in patients, TARDBP mutations are overwhelmingly heterozygous. Homozygosity is thus a severe, arguably non-physiological sensitized background that may exaggerate nuclear RNA mis-splicing and phase defects and alter the relative contribution of cytoplasmic aggregation versus nuclear loss-of-function. In addition, it is not fully clear from this manuscript whether the structures in K181E organoids are bona fide anisosomes as defined in Yu et al. 2021, characterized by HSP70-enriched central liquid cores with TDP-43 shells and similar FRAP and fusion behavior to anisosomes in the DLD1 model. At present, the organoid section is framed as validation of "anisosome-bearing organoids," but the figures in this manuscript mainly show pTDP-43 puncta and total TDP-43 immunostaining, without detailed structural or biophysical characterization. The authors should explicitly compare heterozygous K181E/+ organoids or another heterozygous TARDBP mutant line with homozygous K181E/K181E organoids to assess whether XPO1 inhibition has similar effects in a genotype that more closely resembles patient genetics. They should provide direct evidence that the K181E condensates in organoids are anisosomes through HSP70 core immunostaining, three-dimensional reconstruction, and FRAP measurements, and clarify whether KPT-276 is acting on anisosome-like structures or more generic cytoplasmic aggregates or puncta. Without this, the leap from a DLD1 2KQ cancer cell model to human ALS/FTD-relevant neurons is not convincingly supported.

The reviewer is correct that the use of homozygous K181E organoids generates a homogenous background that is more sensitive for detecting phosphor-TDP43. The goal of the experiment was to test whether XPO1 inhibition mitigates the aggregation of a TDP-43 disease mutant. For this purpose, we believe that our experimental setup is suitable. We agree that we should not extrapolate the result to overemphasize on its disease connections. We will revise the paper to tone down this part.

Regarding the immunostained signals in K181E organoids, we did not report them as anisosomes. As widely documented in the literature, p-TPD-43 is widely used as a marker of

pathological TDP-43 aggregation. P-TDP-43 is enriched in pathological aggregates in human ALS and FTL D patients, colocalized with other aggregation signatures such as ubiquitin and other aggregation prone proteins (PMID: 36008843), and is being used as a diagnostic marker for neurodegeneration (PMID: 31661037). Figure 7A showed that inhibiting nuclear export mitigates the accumulation of p-TDP-43 in mutant tissues. We will revise the subheading and the corresponding text to avoid the confusion.

(3) The title and framing assert that "nuclear export governs TDP-43 phase transitions." However, prior studies such as Pinarbasi et al. 2018 and Duan et al. 2022 indicate that TDP-43 is not a canonical XPO1 cargo and that its export is largely passive, with active nuclear import being the dominant determinant of nuclear localization. The authors cite these studies but still position XPO1 as a central, quasi-direct regulator. The data presented are largely correlative or based on pharmacologic manipulation and overexpression in an overexpression mutant background, with no direct evidence that XPO1 engages TDP-43 in a specific, regulated manner. Even if XPO1 does not engage WT TDP-43, it could still engage the 2KQ variant, which needs to be tested.

We did not conclude or imply the regulation of TDP-43 by XPO1 is direct. In fact, we explicatively mentioned on page 8 that the regulation is likely indirect and mediated by other factors. The sentence reads as "Since XPO1 does not bind TDP-43 directly (Pinarbasi et al., 2018), additional factors likely facilitate XPO1-mediated TDP-43 nuclear egression under this condition." We can revise the part to make it clearer. We will also revise the title and change the framing accordingly.

(4) The XPO1 perturbations yield somewhat confusing phenotypes. XPO1 inhibition using Leptomycin B, KPT-276, and Verdinexor reduces anisosome number and enlarges remaining anisosomes, which remain liquid-like by FRAP recovery and fusion assays and stay nuclear. XPO1 overexpression causes fewer, enlarged puncta, but these are FRAP-impaired (gel-like) and redistribute to the cytoplasm. Thus, both decreased and increased XPO1 activity reduce anisosome number and enlarge puncta, but with opposite phase behaviors and subcellular localizations. The model presented in Figure 5L is relatively qualitative and does not resolve these issues. Moreover, XPO1 inhibition globally impairs nuclear export of many cargos and profoundly alters the nuclear environment, transcription, RNA processing, and chromatin. It is therefore difficult to conclude that the observed effects are specific to TDP-43 phase regulation as opposed to secondary consequences of broad nuclear export blockade.

The reviewer correctly summarizes our data and interpretation: XPO1 loss-of-function and gain-of-function generate opposite phenotypes regarding TDP-43 phase behavior. We agree that additional studies are needed to elucidate the underlying mechanism (e.g. direct or indirect), but we feel that belong to a separate study. We plan to re-test the effect of nuclear export inhibition on the subcellular distribution of WT TDP-43 and the acetylation mimetics. We will also add more discussions about the potential indirect effect of XPO-1 inhibition on TDP-43 phase behavior.

(5) The authors show that anisosome induction depletes nucleoplasmic XPO1 signal and that mCherry-XPO1 can be seen in some TDP-43 puncta. However, antibody penetration into anisosomes is limited, so XPO1 depletion from nucleoplasm could reflect sequestration in the anisosome shell or core, but this is not demonstrated. There is no demonstration of physical interaction, even indirect interaction, between XPO1 and TDP-43 or a defined adaptor, nor identification of a specific mutant of XPO1 that selectively disrupts this putative interaction while preserving other functions. The known TDP-43 NES has been shown to be weak and not a functional XPO1-dependent NES in multiple studies. If XPO1 is acting through an adaptor that recognizes 2KQ or K181E specifically, that by itself would bring into question the generality of the mechanism for wild-type TDP-43.

We agree that our observation does not demonstrate an interaction between XPO1 and TDP-43. As mentioned above, we did discuss that the regulation of TDP-43 by XPO1 is likely indirect. We will revise our paper further to separate any speculative statements from the data and narrow our mechanistic claim.

(6) To support a mechanistic claim that nuclear export governs TDP-43 phase transitions, more targeted evidence is needed. The authors should test whether siRNA knockdown or CRISPR interference of XPO1 in the DLD1 2KQ model reproduces the effects seen with Leptomycin B and KPT-276, including FRAP and fusion phenotypes, and verify on-target effects by rescue with an siRNA-resistant XPO1 construct. They should demonstrate that canonical XPO1 cargos behave as expected under the inhibitor conditions used, as a positive control, and that the concentrations used are not grossly toxic. They should attempt to identify or at least constrain candidate adaptors that might enable XPO1-dependent export of TDP-43 through proteomic analysis of XPO1 co-purifying with 2KQ condensates or loss-of-function studies of candidate adaptors from the siRNA screen. Finally, they should test whether a TDP-43 mutant that cannot bind the proposed adaptor still responds to XPO1 manipulation.

The anisosome enlargement phenotype upon XPO1 depletion was seen in our siRNA screen, which was identified by machine-based image analyses using 6 distinct siRNAs. This, together with the chemical inhibition experiments, convinced us that the phenotype is specifically caused by XPO1 inactivation.

When characterizing the effect of XPO1 inhibition on anisosome dynamics, we preferred chemical inhibitor because the effect is acute, and is therefore, less likely to be caused by secondary effects.

Regarding the inhibitor concentration, a literature survey suggested that 50-200nM of Leptomycin B was commonly used. We chose 200nM to ensure a quick and complete inhibition of XPO1-mediated nuclear export (see Figure 3 in PMID: 9628873). This dose is also well tolerated by our cells, at least during the chosen time window.

We did not propose any specific adaptor that mediates XPO1 interaction with TDP-43. The identification of such adaptor is out of the scope of this study. We will revise our paper to avoid this confusion.

(7) Even with these data, what is currently shown is that global modulation of nuclear export capacity can alter the phase behavior and localization of a highly overexpressed RNA-binding-defective TDP-43 mutant and of K181E in organoids. This is important, but it is weaker than asserting that XPO1 directly governs TDP-43 phase transitions in physiological contexts. The title, abstract, and Discussion should be tempered to reflect that nuclear export is one of several pathways, alongside RNA splicing, translation, and proteostasis, that influence TDP-43 phase states in this model, and that the specific

mechanism and cargo relationship between XPO1 and TDP-43 remain unresolved and may be indirect.

We will revise the title, abstract, and discussion to temper the conclusion.

(8) The authors conclude that cytoplasmic TDP-43 aggregation plays only a modest role in TDP-43 proteinopathies because in homozygous K181E organoids, chronic KPT-276 treatment almost abolishes cytoplasmic pTDP-43 puncta, yet bulk RNA-seq shows only a relatively small fraction of dysregulated genes are rescued. There are several issues with this inference. Relying primarily on pTDP-43 antibody staining to define cytoplasmic TDP-43 aggregation is limiting. pTDP-43 antibodies label only phosphorylated species and may miss non-phosphorylated, oligomeric, or amorphous TDP-43 species that could still be toxic. Different pTDP-43 antibodies vary in epitope accessibility depending on aggregate conformation and subcellular location. More sensitive approaches, such as high-affinity TDP-43 RNA aptamer probes developed by Gregory and colleagues, biochemical fractionation for SDS-insoluble and urea-soluble TDP-43, and filter-trap assays, would provide a more quantitative assessment of cytoplasmic aggregation and its reduction by KPT-276. Without these, it is not safe to assume that cytoplasmic aggregation has been eliminated, as opposed to one antigenic subclass.

We agree with the reviewer that p-TDP-43 may not represent all aggregate species. However, p-TDP-43 antibodies detect the pathologically validated species most tightly associated with TDP-43 proteinopathies. In human ALS and FTLTDP tissues, cytoplasmic inclusions are strongly immunoreactive for phosphorylated TDP-43 (typically S409/410, as used here). Additionally, p-TDP-43 immunohistochemistry is a routine diagnostic criterion in neuropathology. For these reasons, we believe that the observation that inhibition of XPO1 significantly reduces p-TDP-43 is a very significant finding, as it suggests that an improvement in TDP-43 proteinopathy can be achieved by the inhibition of nuclear transport. We plan to revise the text to better explain the significance of p-TDP-43 staining.

(9) The treatment window, spanning from day 87 to 122 with 20 nanomolar KPT-276, may be too late or too mild to reverse entrenched nuclear RNA-processing defects, even if cytoplasmic inclusions are cleared. Once widespread cryptic exon inclusion and alternative polyadenylation misregulation are established, many downstream changes may become self-sustaining or only partially reversible. Moreover, XPO1 inhibition will massively rewire nucleocytoplasmic transport of many transcription factors, splicing factors, and RNA-binding proteins. Thus, the lack of full transcriptomic rescue cannot be cleanly interpreted as evidence that cytoplasmic aggregates are only modest contributors. It may instead reflect that nuclear dysfunction is primary and XPO1 inhibition does not correct, and may even exacerbate, certain nuclear defects.

We agree with the reviewer that the lack of rescue may be caused by technical issues. We will remove the RNAseq data and related texts since it is not essential for our main conclusion.

(10) To support a causal statement about the modest contribution of cytoplasmic aggregates, one would want more direct measures of neuronal health and function, such as cell death, neurite complexity, synaptic markers, and electrophysiology before and after KPT-276, not only transcriptomics. A way to selectively reduce cytoplasmic aggregation without globally inhibiting nuclear export would allow comparison of outcomes.

We will remove the discussion regarding the role of cytoplasmic aggregates in disease.

(11) Given these caveats, the concluding statements that cytoplasmic TDP-43 aggregation is only a modest contributor should be substantially softened. A more defensible interpretation is that in this homozygous K181E organoid model, chronic global XPO1

inhibition reduces pTDP-43-positive cytoplasmic puncta but only partially normalizes the steady-state transcriptome, suggesting that persistent nuclear RNA-processing defects and other pathways continue to drive pathology.

We agree with the review and will revise this part accordingly.

(12) The screens are a major strength but need more rigorous validation for key hits, especially nuclear transport factors. For the siRNA screen, hits are filtered by anisosome number per nucleus, but there is no direct demonstration in the main text that XPO1 or CSE1L knockdown is efficient at the messenger RNA or protein level. For the highlighted genes, Western blot or quantitative polymerase chain reaction validation and phenotypic rescue would strengthen confidence. For small-molecule hits, it is not systematically shown that anisosome modulation is independent of changes in total TDP-43 2KQ expression or gross toxicity. Translation inhibitors are tested for this, but for many other hits, including proteasome, HSP90, and kinase inhibitors, expression and general nuclear structure should be monitored. Given the reliance on anisosome count as a readout, secondary screens that specifically distinguish changes in TDP-43 expression levels, changes in nuclear morphology or cell cycle, and specific changes in anisosome phase behavior, including FRAP and fusion for top hits, would greatly increase interpretability.

For the siRNA screen, each positive hit was confirmed by two rounds of screen with 6 independent siRNAs in total. Although we did not validate the knockdown efficiency due to the large number of hits, we routinely include a positive siRNA control in our study (siRNAdeath), which targets an essential gene. Transfection efficiency was controlled by measuring cell viability after knocking down this essential gene. In addition, the identification of XPO1 as a positive regulator of TDP-43 phase behavior was independently validated by our chemical genetic screens. We feel confident that XPO1 is a key modulator of TDP-43 phase behavior. For chemical treatment experiments, the anisosome fusion phenotypes could be detected as early as 5 h post treatment. Given the short treatment, we do not expect a significant change in protein level or toxicity.

(13) The classification of condensates as liquid versus gel-like or solid is based almost entirely on FRAP recovery or lack thereof. While FRAP is appropriate, interpretations could be made more robust by including half-region-of-interest bleach controls and assessing mobile fractions and recovery kinetics more quantitatively across conditions. Complementing FRAP with other phase-behavior assays such as sensitivity to 1,6-hexanediol, shape relaxation after deformation, and coarsening behavior over longer timescales would strengthen the analysis. At present, some assignments, such as that XPO1 overexpression drives a gel-like transition, are reasonable but somewhat qualitative.

In this study, we described two types of condensates formed by TDP-43 2KQ, one characterized previously as nuclear anisosome and the other as cytosolic puncta in XPO1 over-expressing cells. The two can be clearly distinguished by several features including the subcellular localization, shape, and mobility. We feel that our FRAP data clearly segregate these puncta into two distinctive types of assemblies. The difference in fluorescence recovery rate is huge. The proposed half-region-of-interest bleach is technically challenging for small anisosomes under normal conditions. When they were enlarged by Leptomycin B treatment, we did perform both whole anisosome bleach and partial bleach (Figure 5D, I). Both assays demonstrate that TDP-43 in these enlarged anisosomes is highly mobile.

(14) For the Leptomycin B and KPT-276 experiments in cells and organoids, it would be important to confirm that canonical XPO1 cargo proteins accumulate in the nucleus and that the concentrations used are within a range that is not overtly toxic over the experimental timeframe. Assessing nuclear morphology, chromatin condensation, and general transcriptional activity through global RNA synthesis or key reporter genes

would ensure that observed effects are not secondary to severe global nuclear export collapse.

In Leptomycin B treatment experiments, we carefully chose a dose that was previously validated (see Figure 3 in PMID: 9628873). Based on our DAPI staining, the nuclear morphology appears normal (Figure 5A). Additionally, in cell line-based experiment, the effect of Leptomycin B on anisosomes was detected 6-8 hours post treatment. The change in global protein synthesis should be relatively minor at this time point. In the organoid experiment, the drug dose was determined by a pre-experiment in which the morphology of organoids was evaluated after prolonged treatment with different doses of the inhibitors.

(15) In the organoid section, it is not clear how many independent iPSC clones and organoid batches were used per condition, nor whether batch effects were assessed in the bulk RNA-seq analysis. This should be fully specified and ideally controlled with isogenic wild-type and K181E clones. For transcriptional rescue, it is important to know whether the changes in wild-type organoids treated with KPT-276 are negligible. A direct wild-type comparison with or without KPT-276 is important to disentangle general drug effects from K181E-specific rescue. More detailed quantification of total TDP-43 and pTDP-43 in both nuclear and cytoplasmic fractions, including biochemical fractionation if possible, would strengthen the assertion that KPT-276 specifically reduces cytosolic pTDP-43 aggregates while sparing nuclear TDP-43.

The organoid experiment was performed with two batches per condition. This is to reduce the effect of batch variation. The wildtype cells and K181E mutant are derived from the same genetic background. We will revise the text to clarify these issues. Given the cost of this experiment, we did not include drug-treated wild-type as a control. Given the criticisms by review 1 and 2 on the RNAseq data, we will remove this non-essential data from our revision.

(16) Beyond the core issues above, several additions could greatly enhance the impact. The manuscript currently emphasizes XPO1, but the genetic and chemical data clearly implicate RNA splicing, translation, and proteostasis as equally strong or stronger regulators of TDP-43 phase states. A more integrated model that explains how these pathways intersect, for example, how splicing factor availability, ribosome loading, and proteasome capacity co-govern anisosome nucleation, growth, and hardening, would be valuable.

We agree with the reviewer that these are important directions for future studies. We will include some discussions on a possible model that integrate these factors.

(17) A key unresolved question is whether XPO1 is acting directly on TDP-43, or instead primarily regulates anisosomes by exporting other factors that more proximally control TDP-43 phase behavior. Given that TDP-43 is not a canonical XPO1 cargo and prior work indicates that its nuclear export is largely passive, it seems at least as plausible that XPO1 inhibition alters the nuclear concentration or localization of splicing factors, RNA-binding proteins, chaperones, or other modifiers identified in the screens, and that changes in these proteins secondarily reshape anisosome dynamics. In other words, XPO1 may be exporting a more direct regulator of anisome formation and hardening, rather than exporting TDP-43 itself in a specific, regulated way. The current data do not distinguish between these possibilities. Systematic identification of XPO1-dependent cargos that colocalize with or biochemically associate with anisosomes, combined with targeted perturbation of their nuclear export, would be needed to determine whether the relevant XPO1 substrate in this system is actually TDP-43 or an upstream modulator of its phase behavior.

The reviewer raises an important point. We did include some discussions along this line in our paper. We can add more to further clarify this issue. Again, as mentioned in the original

draft, we did not conclude there is an interaction between TDP-43 and XPO1.

(18) Testing whether identified modifiers converge on nuclear TDP-43 concentration would be informative. Since phase separation is concentration-dependent, measuring nuclear versus cytoplasmic TDP-43 levels across key perturbations, including splicing inhibition, translation inhibition, proteasome inhibition, HSP90 inhibition, and XPO1 modulation, would help determine whether modifiers mainly work by changing nuclear TDP-43 concentration or by altering interaction networks and the material properties of condensates.

We will measure the nuclear TDP-43 concentration in our imaging experiments and add the data to a revised version.

(19) Examining other ALS-relevant RNA-binding proteins would be valuable. Given the role of XPO1 and other hits, it would be informative to briefly test whether similar principles apply to FUS, hnRNPA1, or other ALS-relevant RNA-binding proteins in the same cellular context, to argue for generality versus TDP-43-specific idiosyncrasies of the 2KQ system.

We agree that this is an important issue but we feel the proposed experiments are beyond the scope of the study.

(20) The Introduction sometimes implies that anisosomes are common and well-established intermediates en route to pathology. It would be helpful to more clearly state that, to date, anisosomes are primarily observed in overexpression and mutant systems and have not yet been unequivocally demonstrated in human patient tissue. The link between PDGFR β , PAK4, GSK-3 β , and YAP and TDP-43 phase dynamics is intriguing but only briefly mentioned. The authors should either expand on this or tone down the emphasis in the Results section.

We will revise the introduction accordingly.

(21) In the organoid methods, the authors should consider clarifying whether doxycycline is continuously used, which might alter TDP-43 expression and nuclear transport in a non-negligible way.

The organoid model does not involve protein overexpression or doxycycline treatment. We measured endogenous p-TDP-43. We will revise to paper to avoid the confusion.

(22) For statistical methods, it would be beneficial to indicate whether multiple-comparison corrections were applied for the many FRAP, anisosome count, and size comparisons beyond DESeq2 internal corrections for RNA-seq.

We will add this information to the figure legends during revision.

(23) Some figure legends could more clearly indicate whether the images shown are single z-planes or maximum intensity projections and how the thresholding for anisosome detection was performed.

We will revise the figure legends to include this information. As for anisosome detection, because they are so obvious, standard thresholding was sufficient to identify them.

(24) In its current form, the manuscript contains an impressive set of screens and some nicely executed imaging of TDP-43 condensates, highlighting nuclear export among other pathways as a modulator of TDP-43 phase behavior. However, the physiological relevance is undercut by heavy reliance on an acetylation-mimetic, RNA-binding-defective TDP-43 mutant and a homozygous K181E organoid model. The mechanistic link between XPO1 and TDP-43 remains largely inferential and partly at odds with prior work. The

conclusion that cytoplasmic TDP-43 aggregation is only a modest contributor to disease is not firmly supported by the available data.

We agree with the reviewer that the strength of the study is our unbiased approach that identify pathways capable of modulating TDP-43 phase separation behavior. We will revise our paper to carefully discuss the potential physiological relevance of our study and tone down some mechanistic conclusions, as suggested by the reviewer.

(25) With substantial additional mechanistic work, particularly around XPO1, rigorous validation in more physiological TDP-43 contexts, more sensitive detection of cytoplasmic TDP-43 aggregates, and a tempering of the central claims, this study could make a meaningful contribution to understanding how nucleocytoplasmic transport and other cellular pathways influence TDP-43 phase transitions and aggregation. The work should be reframed as an important screening study that identifies nuclear export as one among several cellular processes that modulate TDP-43 phase behavior in a model system, rather than as a definitive demonstration that nuclear export governs pathological TDP-43 aggregation in disease.

We will reframe the study as an important screening study that identifies nuclear export among several other pathways as modulators of TDP-43 phase behavior.

Reviewer #2 (Public review):

Summary:

This manuscript addresses an important and timely question in TDP-43 biology by systematically identifying regulators of TDP-43 anisosome formation, with a particular focus on nuclear export via XPO1. Using a combination of unbiased chemical screening, genetic perturbation, and advanced imaging approaches, the authors propose that inhibition of nuclear export modulates the abundance and biophysical properties of TDP-43 anisosomes. The study is conceptually innovative and has potential relevance for neurodegenerative diseases characterized by TDP-43 pathology. However, significant concerns regarding experimental controls, reporting transparency, and model translatability currently limit the strength of the conclusions and the interpretability of several key findings.

We thank the reviewer for acknowledging the significance and innovation of our study.

Strengths:

(1) The study employs an unbiased, hypothesis-free compound screen to identify regulators of TDP-43 anisosome formation, which is a major strength and reduces confirmation bias.

(2) The authors combine chemical and genetic screening approaches, providing orthogonal validation of key pathways and increasing confidence in the biological relevance of top hits.

(3) The focus on biophysical properties of TDP-43 assemblies, assessed through imaging and FRAP, moves beyond simple presence/absence of aggregates and provides mechanistic insight into the biophysical states of TDP-43.

(4) The use of multiple experimental modalities, including live-cell imaging, FRAP, pharmacological perturbation, and transcriptomic analysis, reflects a technically sophisticated and ambitious study design.

(5) The authors attempt to extend findings beyond immortalized cancer cell lines by incorporating organoid models, demonstrating awareness of disease relevance and

translational importance.

Overall, the manuscript is clearly written and logically structured, making complex experimental workflows accessible and the central hypotheses easy to follow.

Weaknesses:

Despite its strengths, the manuscript has several major limitations that affect data interpretation and confidence in the conclusions.

(1) Lack of appropriate controls for overexpression experiments:

A central concern is the absence of proper controls for TDP-43 and XPO1 overexpression. Prior studies (including those cited by the authors, Archbold et al.2018) show that overexpression of WT TDP-43 alone is toxic to neurons. Thus, the experimental system itself may induce anisosome formation independently of the mechanisms under study. Similarly, XPO1 overexpression lacks a suitable control (e.g., mCherry alone or mCherry fused to a protein known to be independent of TDP-43). The near-complete colocalization of XPO1 with TDP-43 anisosomes upon overexpression raises the possibility that these structures reflect non-physiological protein accumulation rather than regulated assemblies.

As mentioned in our response to reviewer 1, point 1, we will add more discussion regarding the use of acetylation mimetics in our study. We agree with the reviewer that these large puncta (both anisosomes and gel-like structures) likely resulted from TDP-43 overexpression. Nevertheless, in a titration experiment done by Yu et al. 2020 (PMID: 33335017), they showed that ectopic TDP-43 undergo demixing even at concentrations lower than endogenous TDP-43, although the demixed puncta were very small. Their result suggested that overexpression per se does not change TDP-43 phase behavior, only enlarging the demixed TDP-43 structures. This is necessary for our screen and imaging-based characterization. We will revise the text to clarify this point.

For XPO1, we did include mCherry alone control in the study but due to space limit in Figure 5, we did not include it. We can put the data in a Supplementary Figure during revision.

(2) Insufficient experimental and analytical transparency:

The manuscript frequently lacks clear reporting of experimental details. In multiple figures, the stated number of independent experiments does not match the number of data points shown, making it difficult to assess statistical validity. Concentrations used in the compound screen are not clearly defined, nor is it stated whether multiple concentrations were tested. It is unclear how many wells, cells, or independent cultures were analyzed. The criteria used to reduce 1,533 screening hits to 211 candidates via STRING analysis are not explained. Knockdown and overexpression efficiencies are not reported.

We apologize for these omissions. We will add more experimental details to the figure legends and the method part. For the imaging experiments, data points reflect randomly selected individual cells imaged in 2-3 independent biological repeats. For chemical screens, we screened against NCATS libraries first at top concentration (10 mM) to ensure inhibitory efficacy for all compounds. In the follow-up study, we validated the top hits using a series of concentrations, as shown in Figure 1B.

We will explain the STRING analysis in more detail. We did not check XPO1 knockdown efficiency in high through-put screens (HTS) for several reasons. Firstly, the large number of positive hits makes it impossible to check knockdown efficiency for all these hits. Secondly, the effect of XPO1 knockdown on anisosomes was seen with 6 different siRNAs in two rounds

of screens. Thirdly, in the HTS protocol, we routinely included a transfection control (siRNAdeath) to indicate high transfection efficiency. We would only process the data if siRNAdeath control killed > 90% of the cells.

(3) RNA-seq concerns:

The RNA-seq experiments are particularly problematic. The number of biological replicates per condition is not stated, and heatmaps suggest that only one sample per group may have been used, which would preclude statistical analysis. No baseline comparison between WT and mutant TDP-43 is shown. Given that TDP-43 is an RNA-binding protein, splicing analyses would be far more informative than gene expression alone, yet no splicing data are presented. Moreover, nuclear retention of TDP-43 does not preclude nuclear aggregation, which may still impair its splicing function.

We apologize for the lack of clarity regarding the RNA-seq design. For each condition, organoids of two independently differentiated batches were treated in triplicate. We pooled the organoids of the same treatment from the two batches to reduce the impact of batch variation.

Given the criticisms from both reviewer 1 and 2 on the limitation of the RNAseq study, we plan to remove this data from the revised manuscript.

(4) Limited translatability to neuronal biology:

All anisosome analyses are performed in a cancer cell line, raising concerns about relevance to post-mitotic neurons. While organoids are used as a secondary model, the assays performed do not overlap with those used in cancer cells, making it difficult to assess whether anisosome-related mechanisms are conserved. Neuronal toxicity, a critical outcome given known TDP-43 biology, is not assessed. Prior work has shown that WT TDP-43 overexpression alone is toxic to neurons, yet this is not addressed.

We agree with the reviewer that the model used in this study is not directly relevant to neurodegeneration. However, as pointed out by the reviewer, neurons are much more sensitive to TDP-43-associated toxicity. By contrast, the cell line used in this study can tolerate TDP-43 overexpression with no detectable cytotoxicity. This feature makes it feasible to evaluate how different cellular processes modulate TDP-43 phase behavior without the confounding effect from toxicity. The fact that TDP-43 expression was induced for a short period of time also help minimize the impact of toxicity. Notably, the processes identified by our screens are all house-keeping pathways that is present in neurons. Thus, we believe that the reported findings are likely applicable to neurons, though we will revise our paper to make sure that we don't overstate the clinical relevance of our work.

(5) Conceptual and interpretational gaps:

The authors quantify anisosome number but also report conditions in which anisosome number decreases while size increases. The biological interpretation of larger anisosomes is not discussed, and whether this reflects improvement or worsening of pathology is unclear. Compounds targeting the same mechanism (e.g., nuclear export inhibition) are inconsistently used across experiments (KPT compounds, verdinexor, leptomycin B), raising concerns about reproducibility. In organoids, the experimental paradigm shifts to long-term treatment (35 days vs. 16 hours), further complicating interpretation.

As pointed out by the reviewer 1 in point 4 above, we do not have evidence to establish a convincing correlation between the size of anisosomes and clinical phenotypes. Regarding the use of different drugs for different experiments, the initial screen identified KPT and VerdineXor because Leptomycin B was not in our library. In the follow-up studies, we

switched to Leptomycin B because 1) it is commercially available; 2) it is highly potent and specific; 3) it was more commonly used as inhibitors of XPO1 according to the literature. However, for the organoid study, we had to switch back to KPT because of the toxicity issue associated with long-term application of Leptomycin B.

(6) Overinterpretation of rescue effects:

Although the authors state that they aim to test whether nuclear export inhibition rescues neuronal defects, no functional neuronal readouts are provided (e.g., viability, morphology, axon outgrowth, or electrophysiological measures). RNA-seq alone is insufficient to support claims of rescue.

Our interpretation of the RNA-seq data was that the rescue effect by nuclear export inhibition was limited and likely insignificant. Given that this negative data is not conclusive, we will remove it from the revised manuscript.

(7) Finally, the model does not appear to exhibit cytosolic TDP-43 aggregation at baseline. It remains unclear whether longer induction would produce cytosolic gel-like assemblies and whether these would be prevented by nuclear export inhibition. Long-term data are shown only in organoids, yet anisosome formation is not assessed there.

The expression system used in the study reaches a steady state after 48 h of induction. At this point, we did not observe any gel-like structures. We can clarify this point during revision.

Reviewer #3 (Public review):

Summary:

TDP-43 proteinopathy is broadly found in neurodegenerative diseases. This manuscript investigates how nuclear export influences the biophysical properties of TDP-43. The authors use a combination of chemical screening and genome-wide siRNA screening to identify pathways that modulate TDP-43 liquid-to-solid transitions. Overall, the study employs a broad array of approaches and addresses an important question in TDP-43 pathobiology. The identification of nuclear export as a central regulator is compelling and conceptually aligns with the emerging view that TDP-43 nucleocytoplasmic trafficking is a major defect in neurodegeneration.

Strengths:

This work integrates chemical and genetic screening to identify novel modifiers. The candidates were validated in both reporter cell lines and iPS-differentiated organoids. The findings support the nucleocytoplasmic transport is important for the biophysical properties of TDP-43.

We thank the reviewer for acknowledging the significance and strength of our study.

Weaknesses:

The mechanisms underlying the connection between nuclear export and phase transition need further clarification. Broader consequences of XPO1 inhibition are not addressed.

We agree that our study does not address how nuclear export inhibition affect TDP-43 phase behavior. As discussed in the paper, we proposed that the effect of nuclear export inhibition on TDP-43 phase separation is likely indirect. The most likely scenario is that inhibition of nuclear export changes the nuclear environment over time, which affects TDP-43 phase separation. We have tried to isolate nuclear extracts from control and LMB-treated cells and used mass spec to identify proteins that are differentially present in the nucleus. However, knockdown of the identified top candidates did not abolish LMB-induced phase alteration.

Considering our observation that RNA splicing is another modulator of TDP-43 phase behavior, it is possible that it is the combined change of RNA and protein composition in the nucleus that alters TDP-43 phase behavior. However, defining the mechanism would require substantial work that is beyond the scope of the current study.

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